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(54) **ADAPTER SYSTEM FOR MULTI-FIBER
MECHANICAL TRANSFER TYPE FERRULE**

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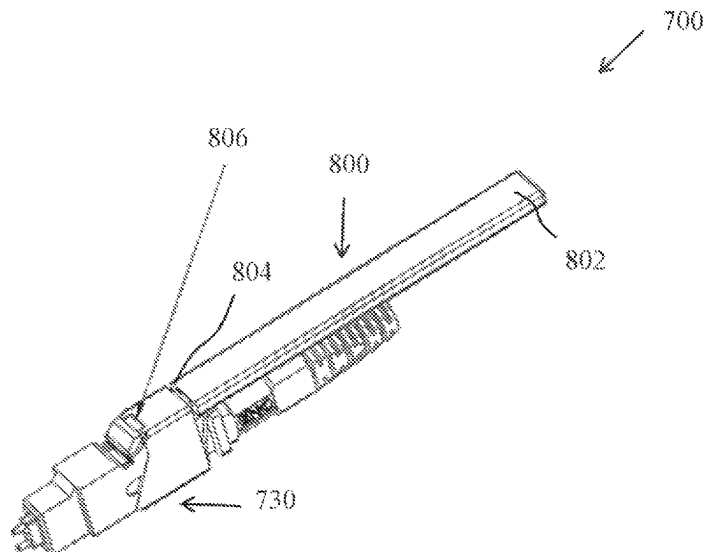
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(57) **ABSTRACT**

Devices and methods for connecting optical fibers are provided. In some embodiments, connectors and adaptors for two-fiber mechanical transfer type ferrules are disclosed. In some embodiments, MT connectors, such as simplex, duplex, and quad micro-MT adaptors are disclosed. In some embodiments, MT adaptors, such as simplex, duplex, and quad adaptors are disclosed. In some embodiments, optical fiber cables that modularly coupled with at least one optical fiber connector, adaptor, and other optical fiber cable the cable is configured to provide a remote release from an adaptor receptacle.

6 Claims, 25 Drawing Sheets



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Exhibit A-3—U.S. Pat. No. 11,733,466 Invalidity Claim Chart, Jun. 21, 2024, 49pgs.

Exhibit A-4—U.S. Pat. No. 11,733,466 Invalidity Claim Chart, Jun. 21, 2024, 37pgs.

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Exhibit B-2—U.S. Pat. No. 11,808,994 Invalidity Claim Chart, Jun. 21, 2024, 64pgs.

Exhibit B-3—U.S. Pat. No. 11,808,994 Invalidity Claim Chart, Jun. 21, 2024, 105pgs.

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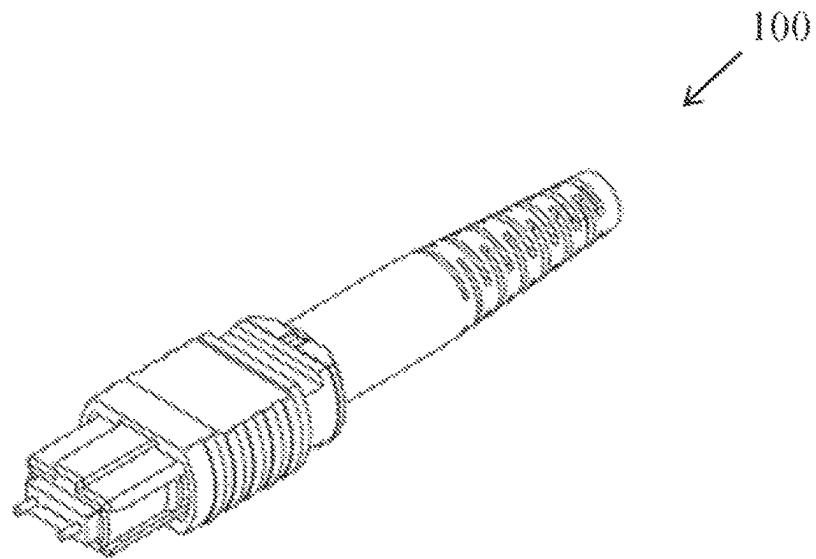


FIG. 1A
(Prior Art)

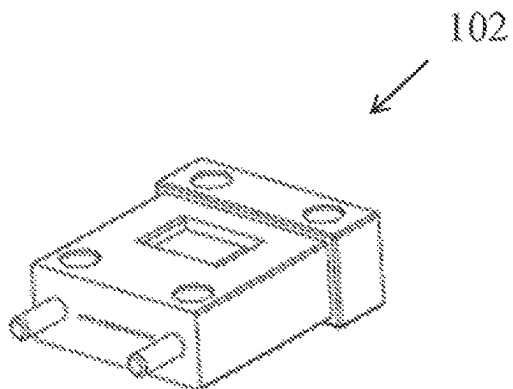


FIG. 1B
(Prior Art)

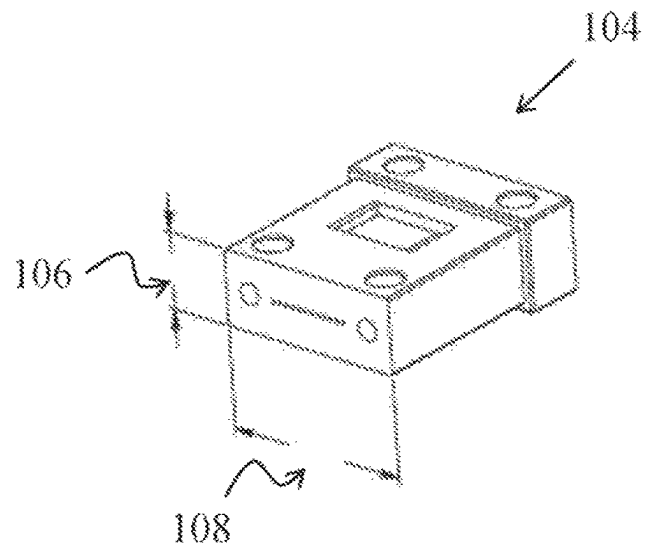


FIG. 1C
(Prior Art)

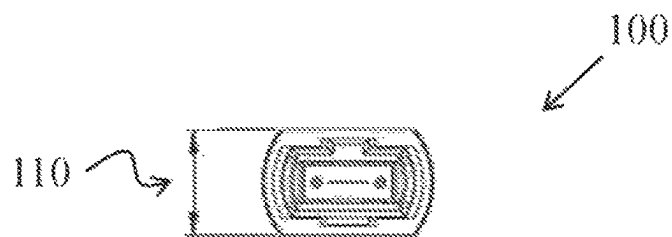


FIG. 1D
(Prior Art)

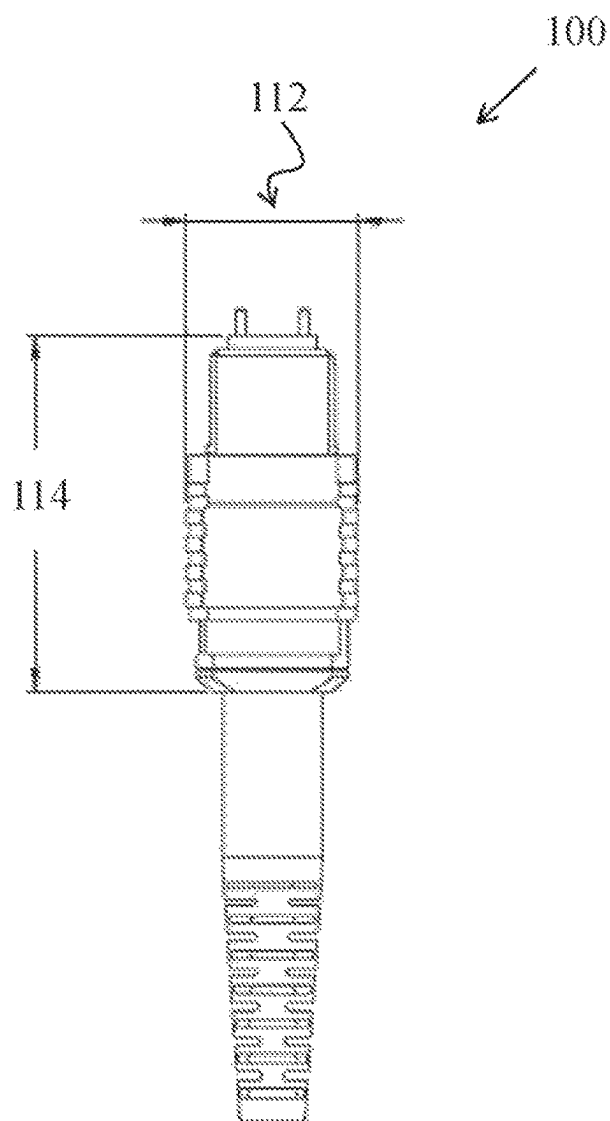


FIG. 1E
(Prior Art)

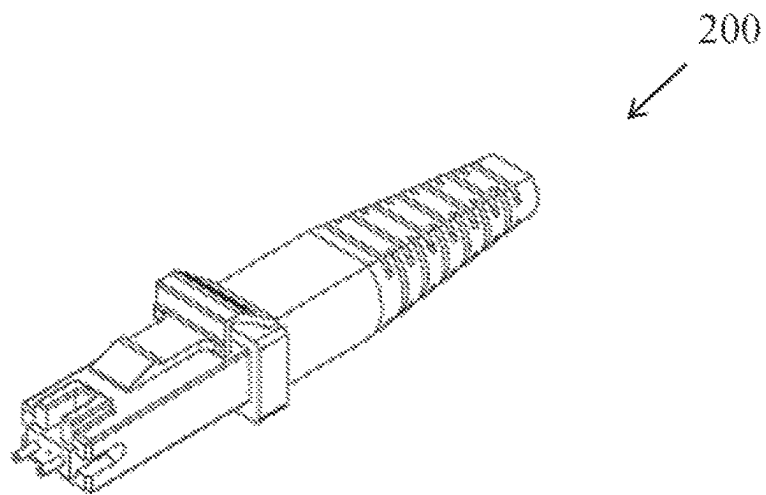


FIG. 2A
(Prior Art)

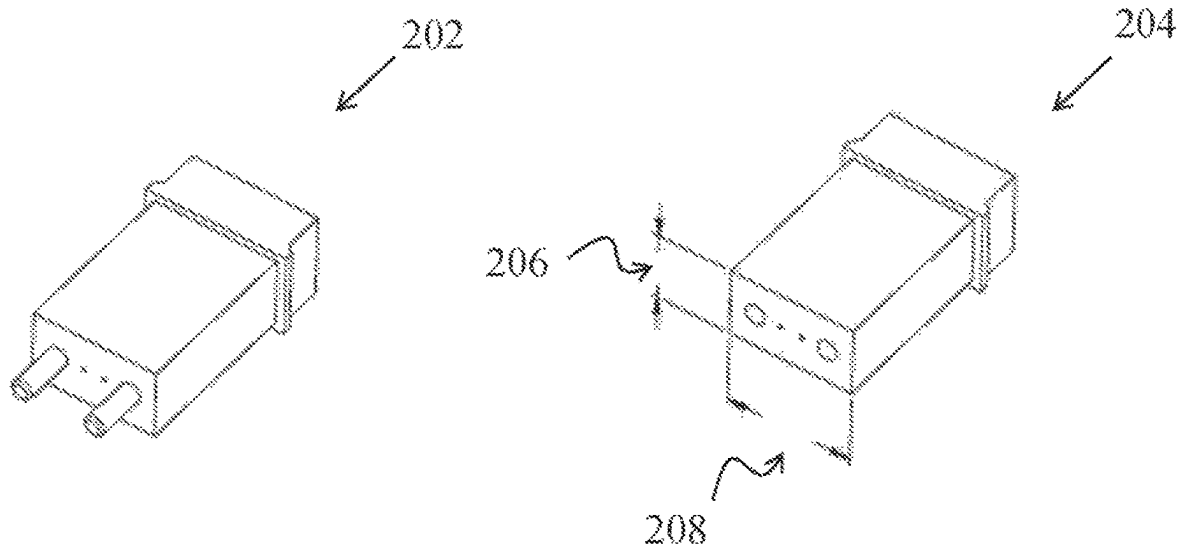


FIG. 2B
(Prior Art)

FIG. 2C
(Prior Art)

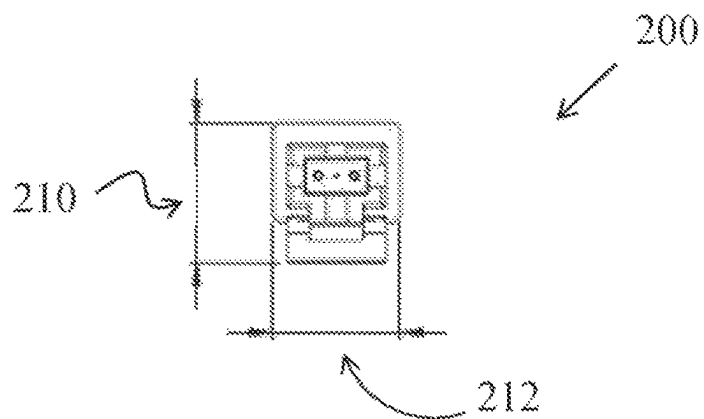


FIG. 2D
(Prior Art)

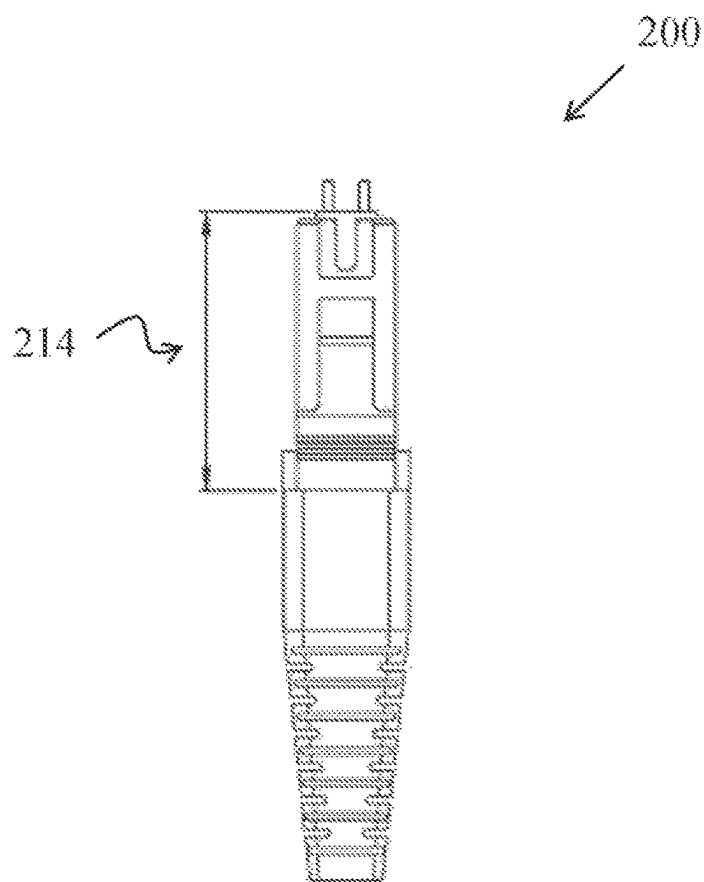


FIG. 2E
(Prior Art)

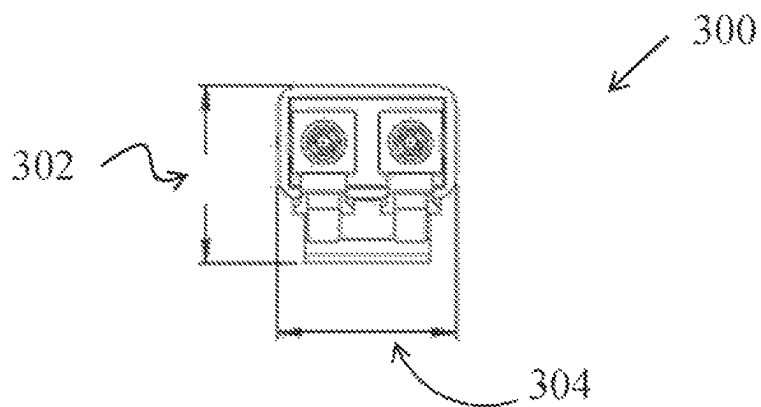


FIG. 3A
(Prior Art)

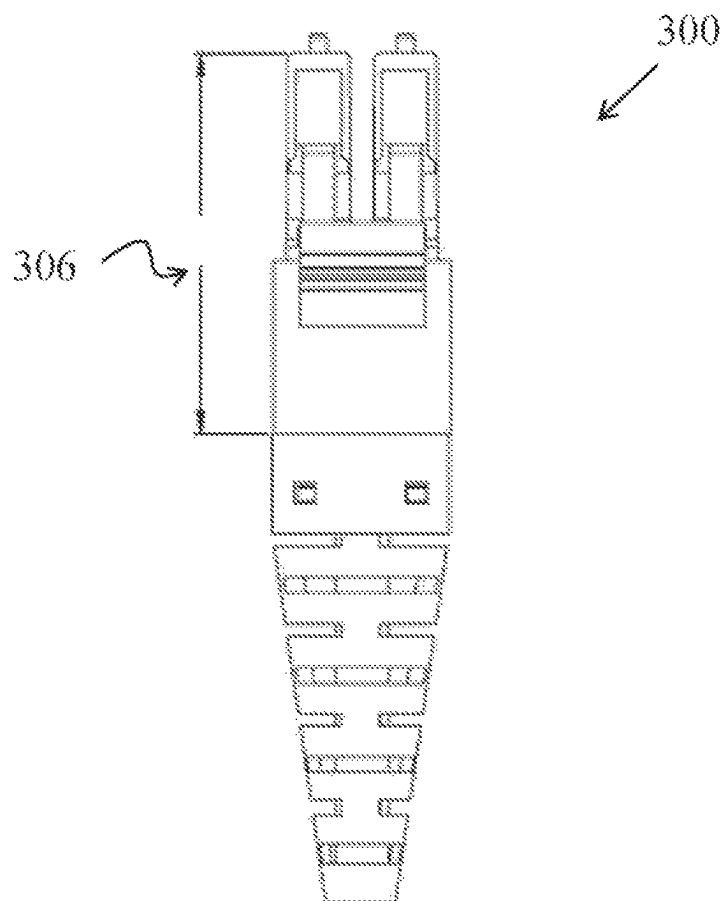


FIG. 3B
(Prior Art)

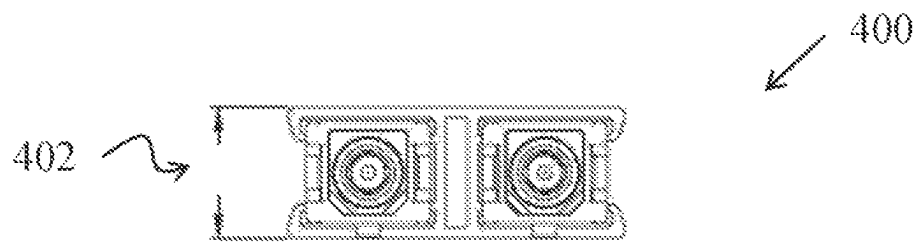


FIG. 4A
(Prior Art)

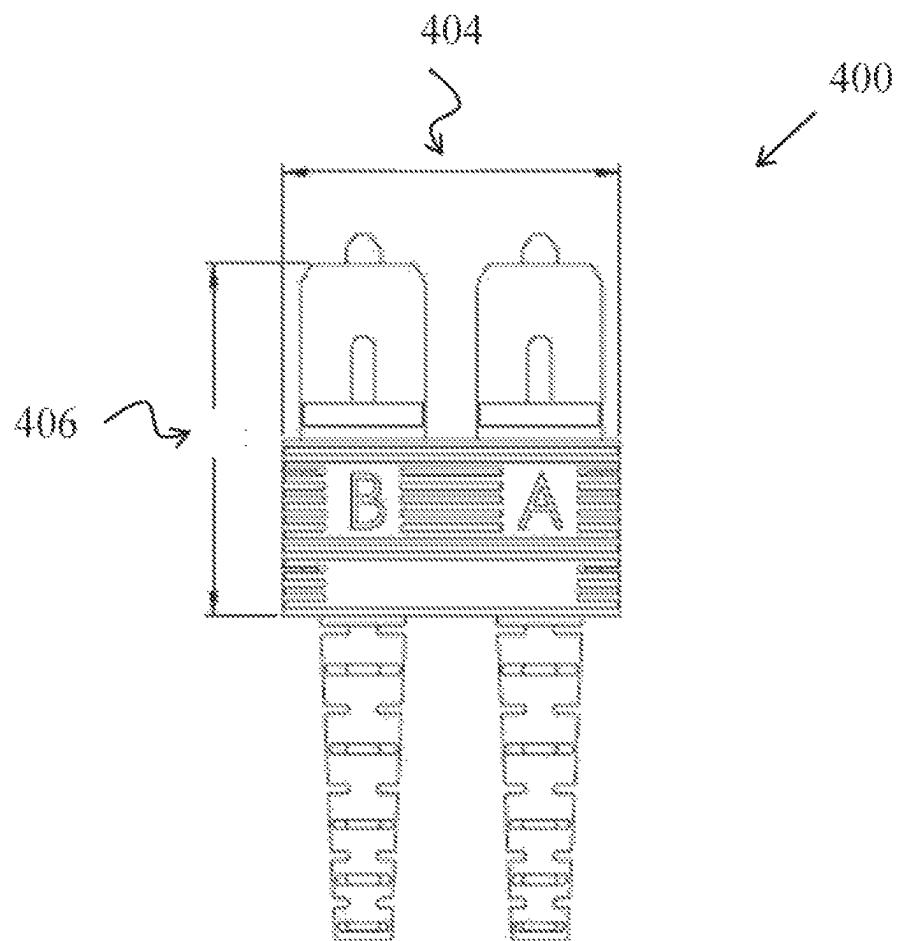


FIG. 4B
(Prior Art)

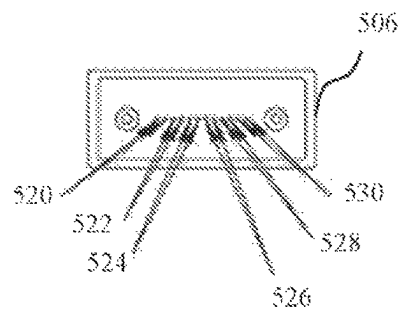


FIG. 5A
(Prior Art)

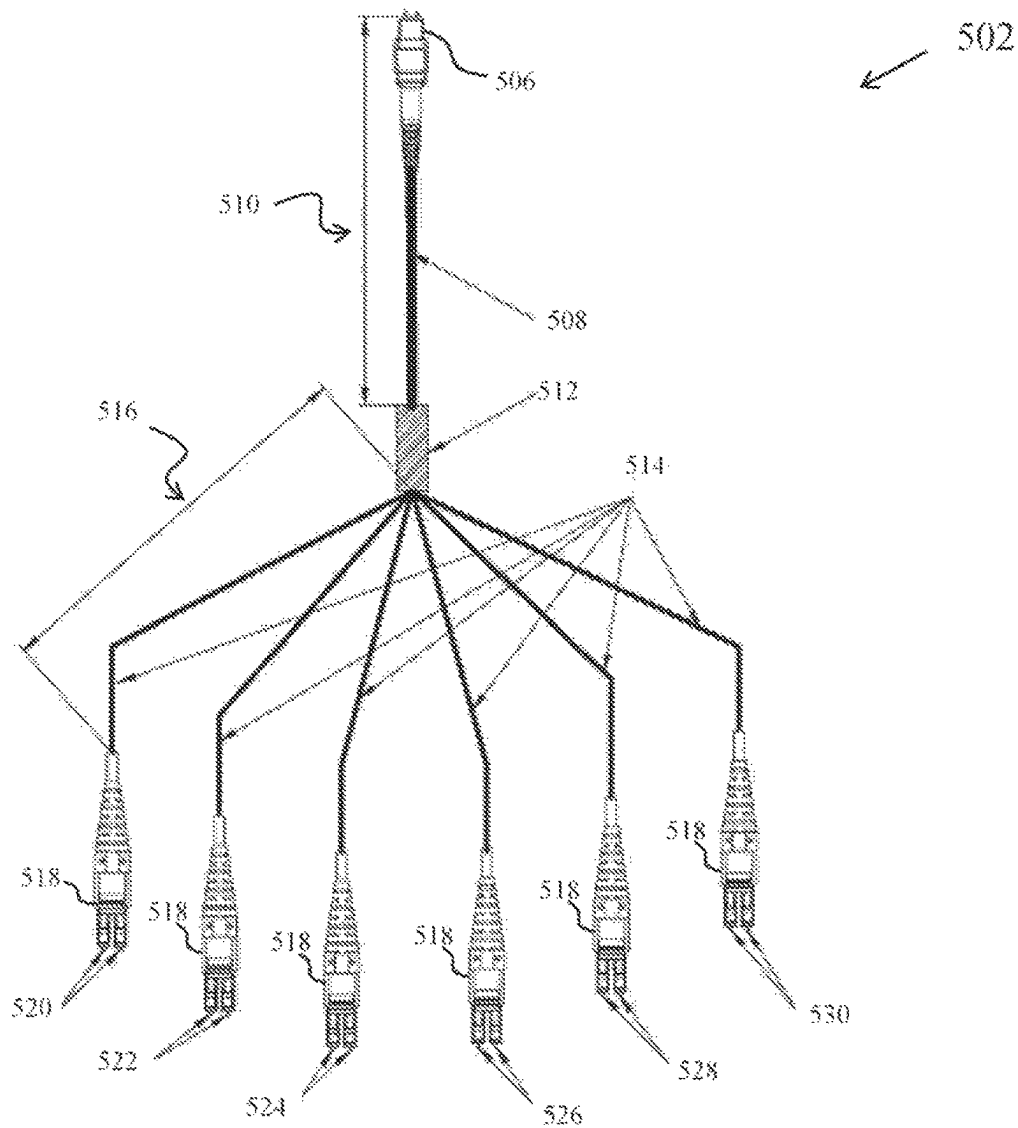


FIG. 5B
(Prior Art)

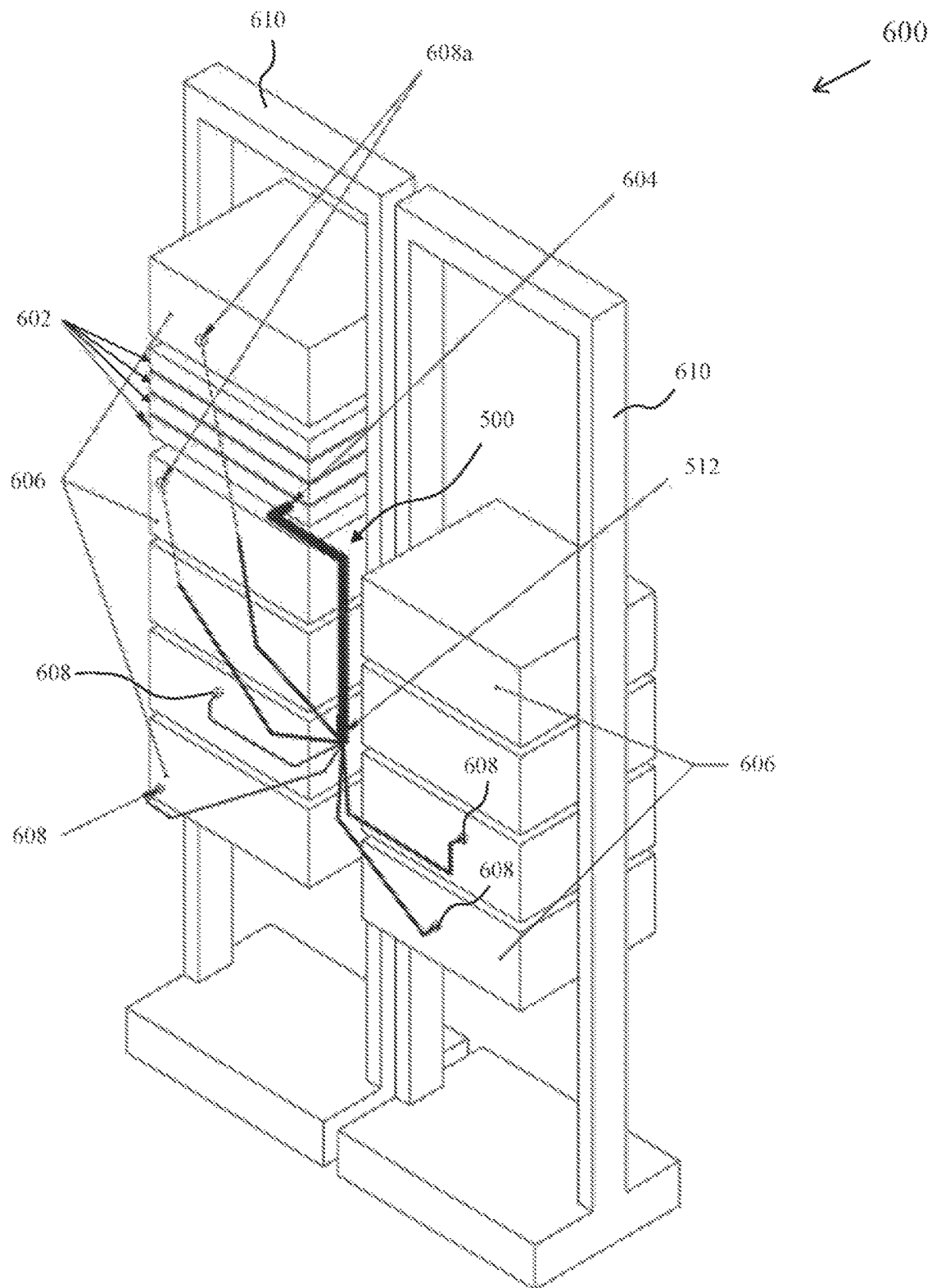


FIG. 6
(Prior Art)

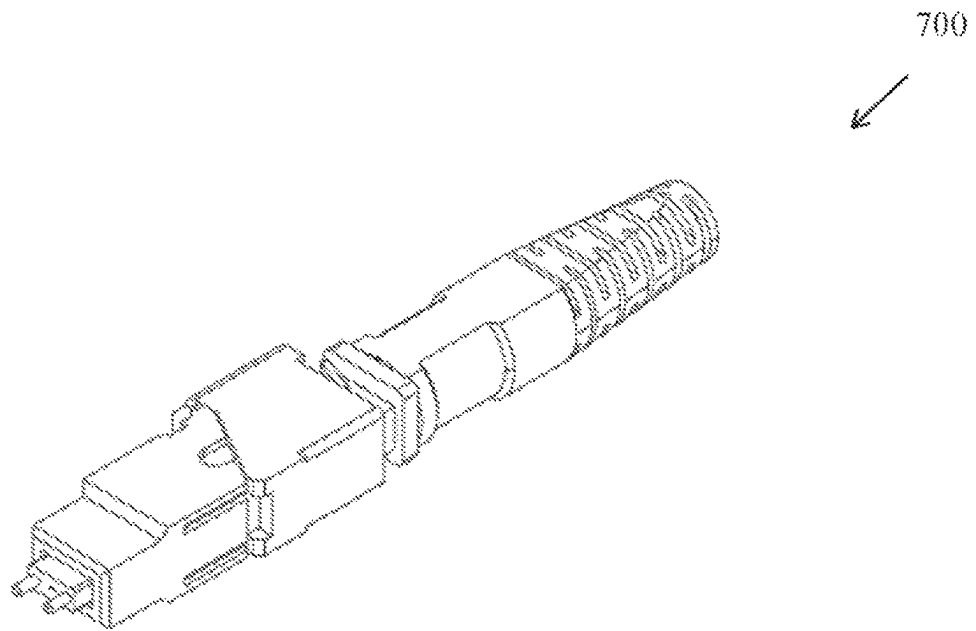


FIG. 7A

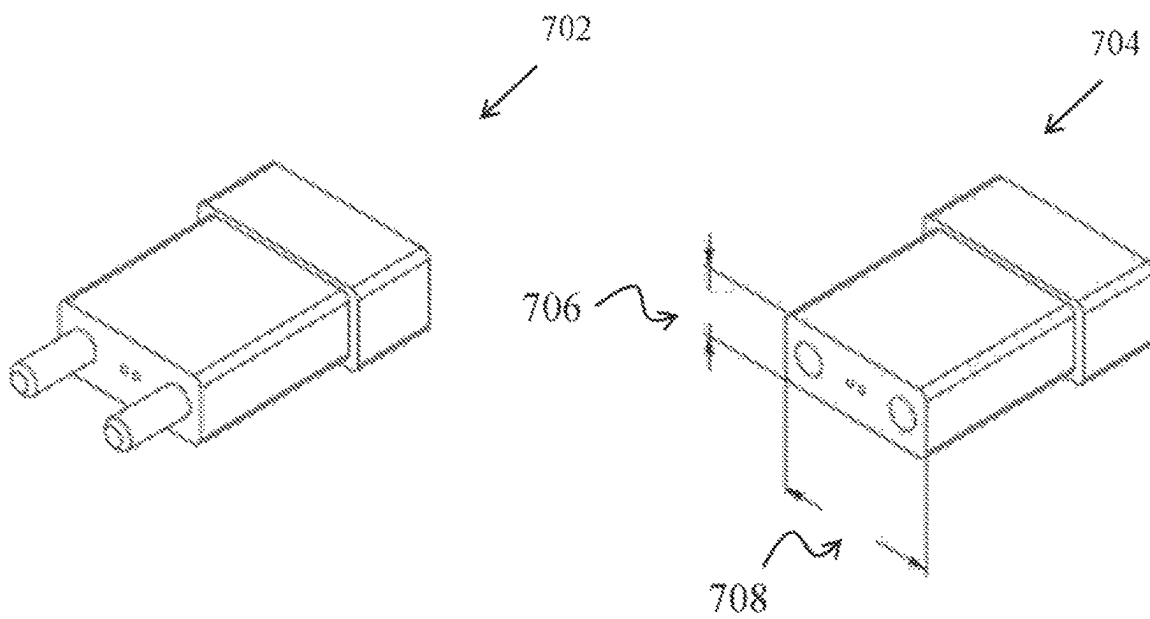


FIG. 7B

FIG. 7C

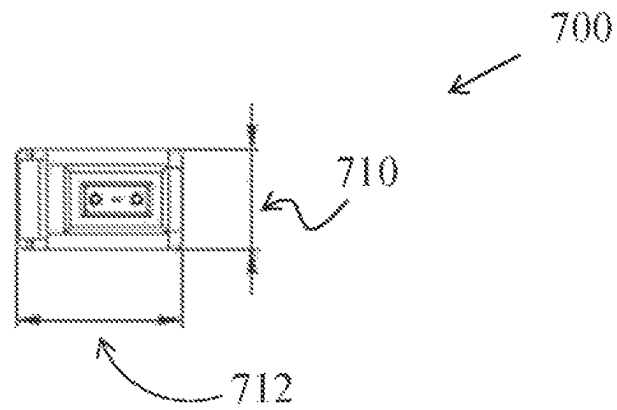


FIG. 7D

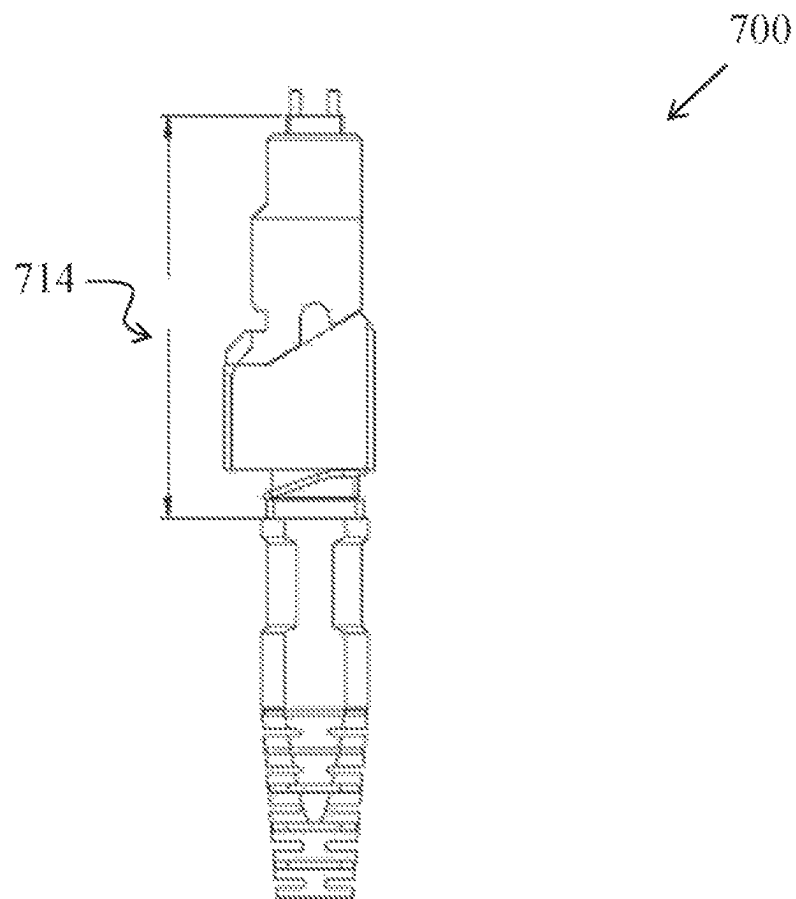


FIG. 7E

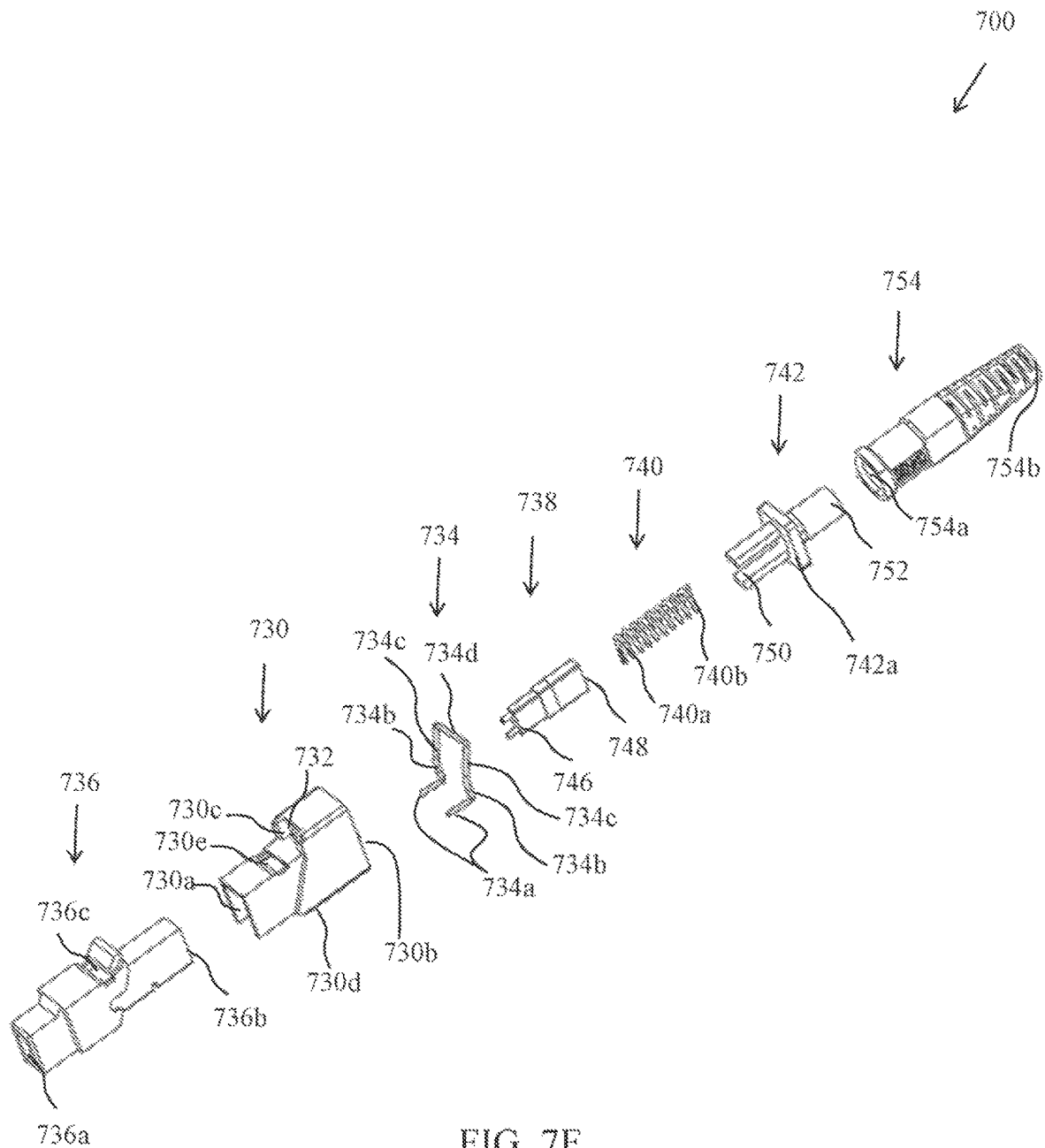


FIG. 7F

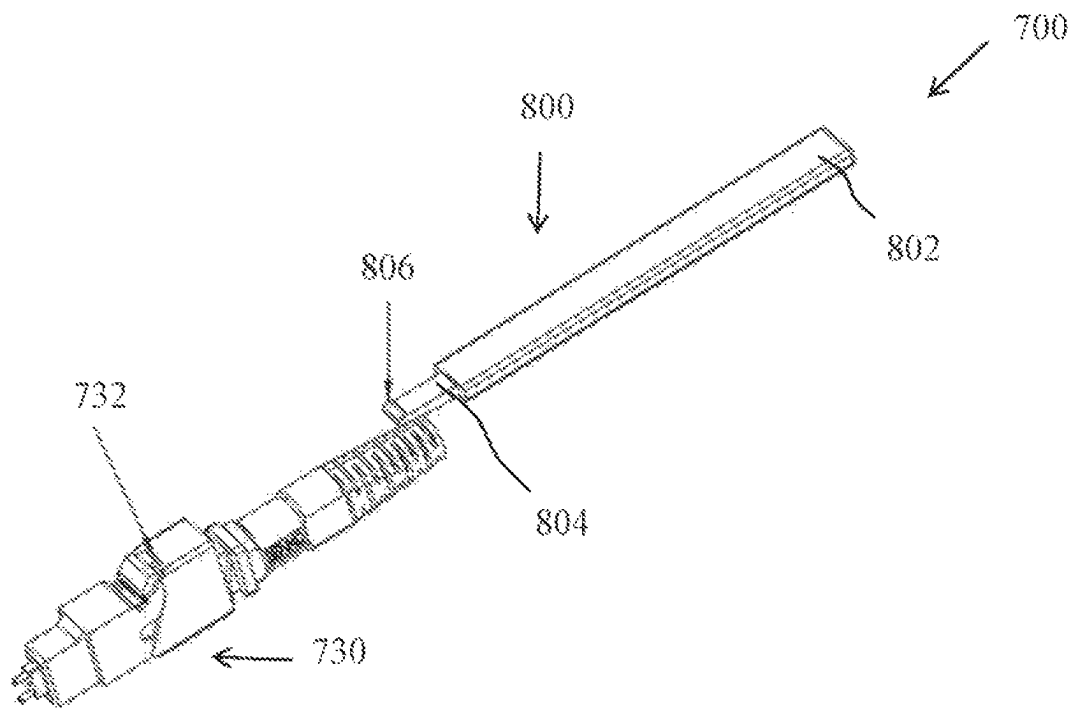


FIG. 8A

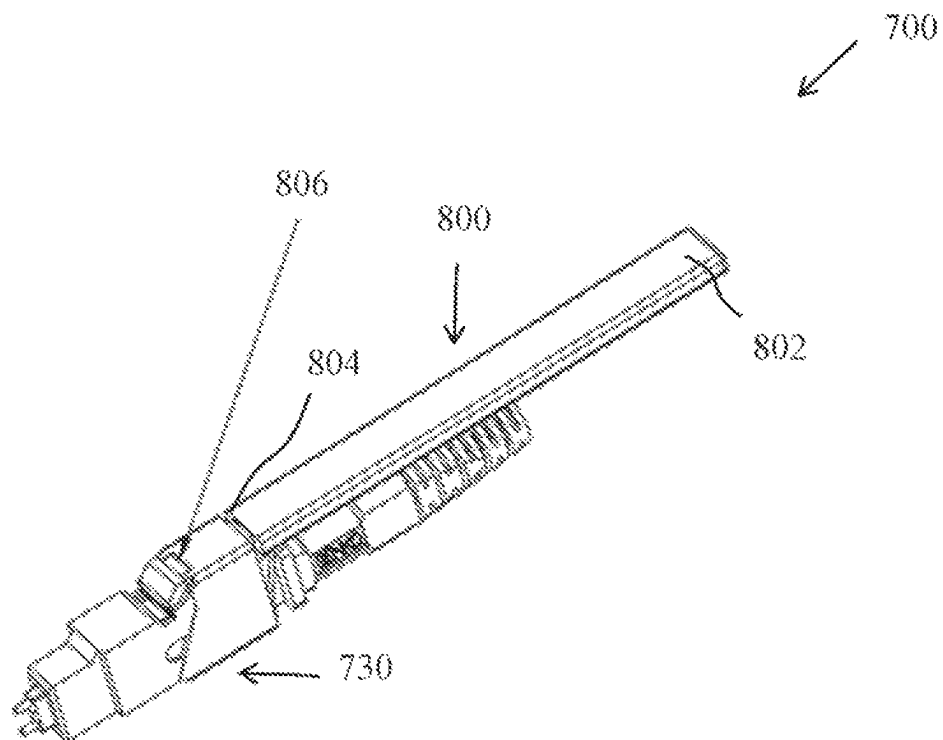


FIG. 8B

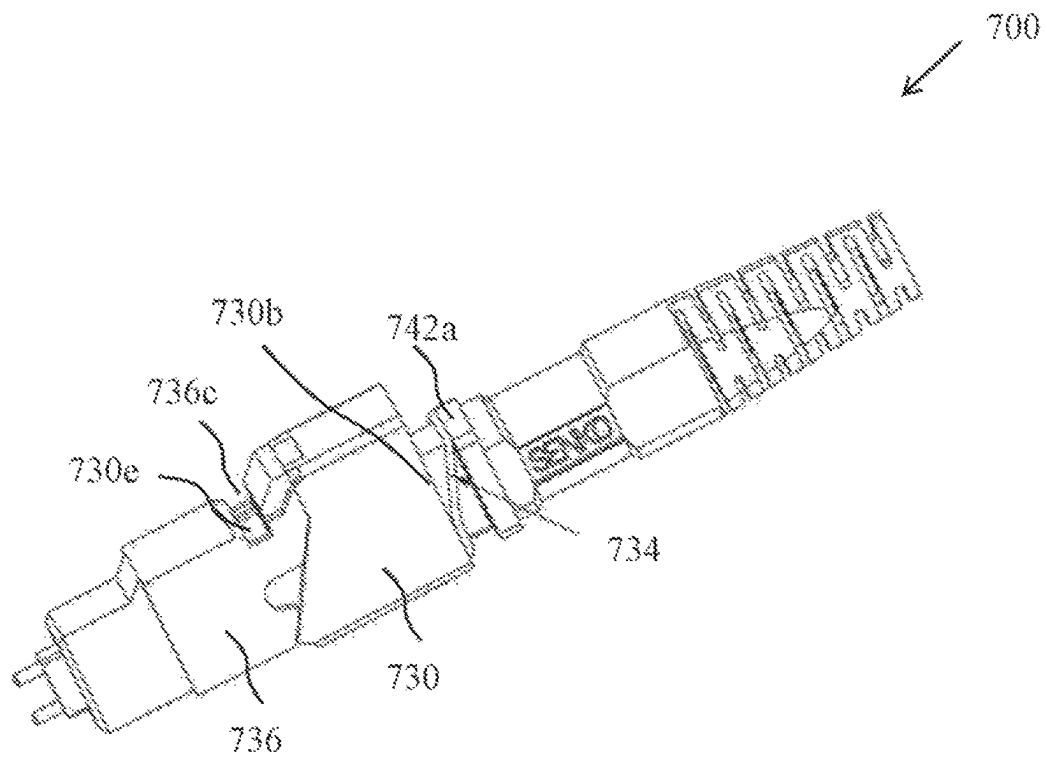


FIG. 9A

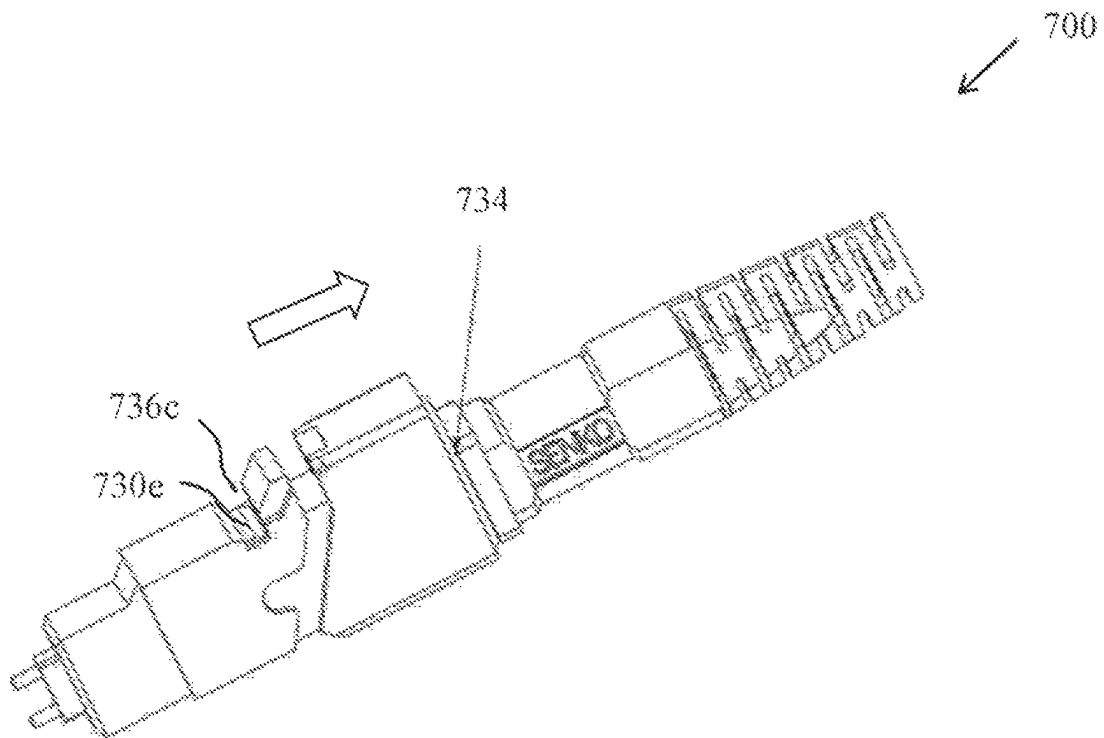


FIG. 9B

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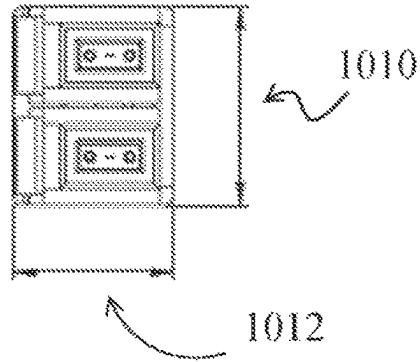



FIG. 10A

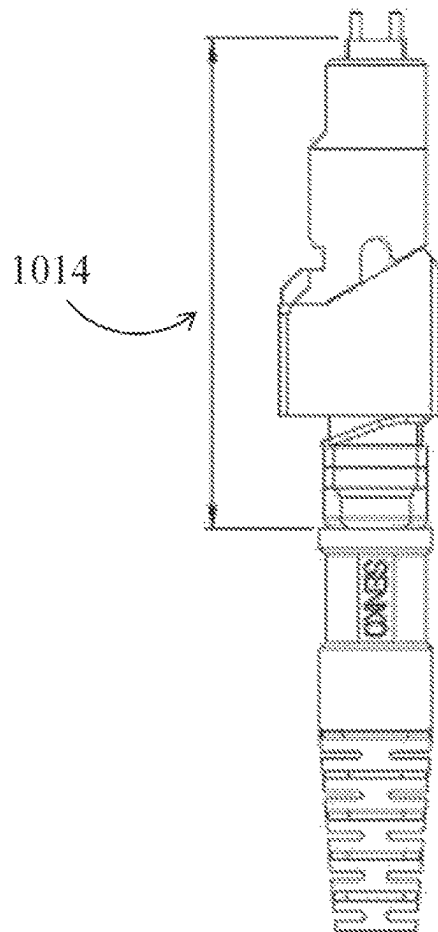


FIG. 10B

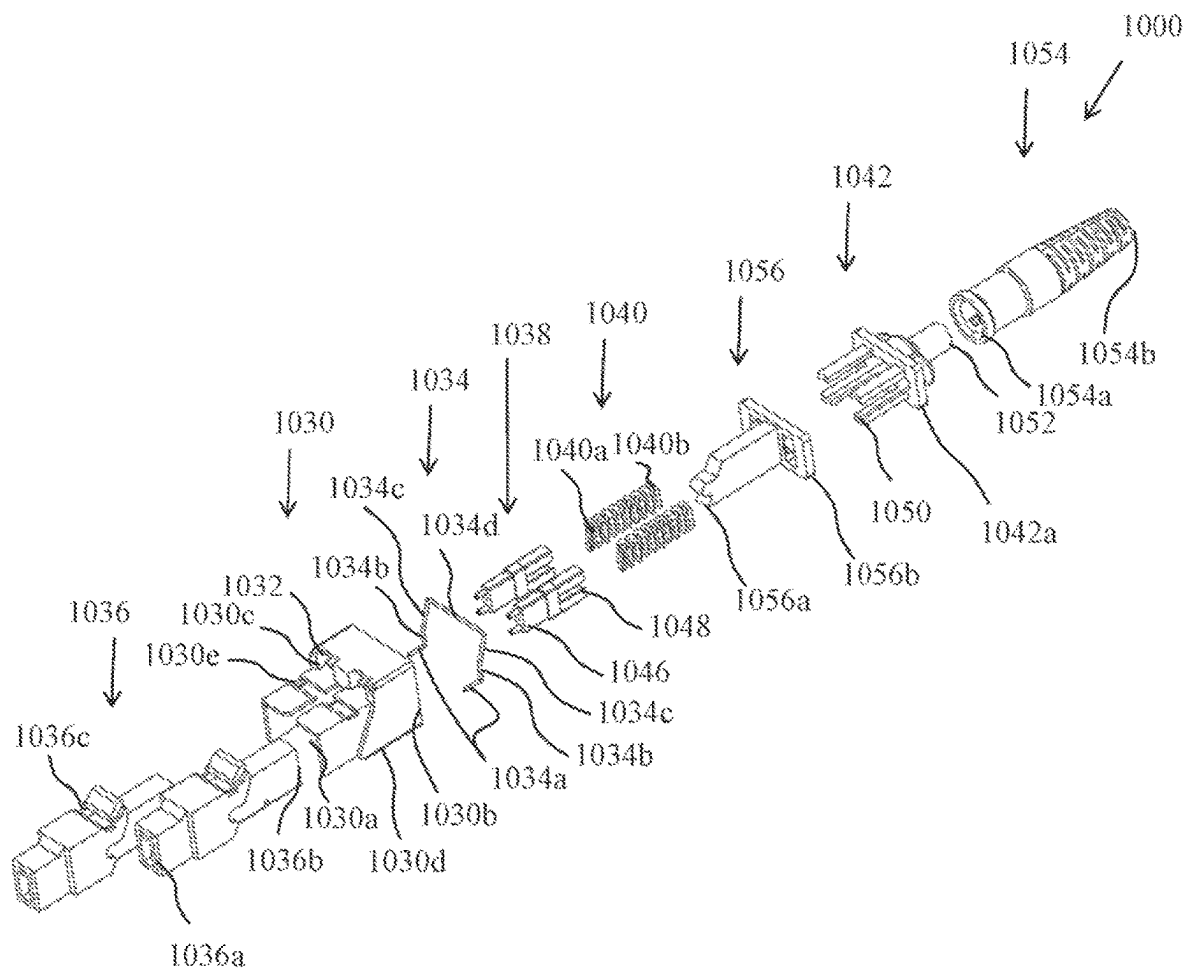


FIG. 10C

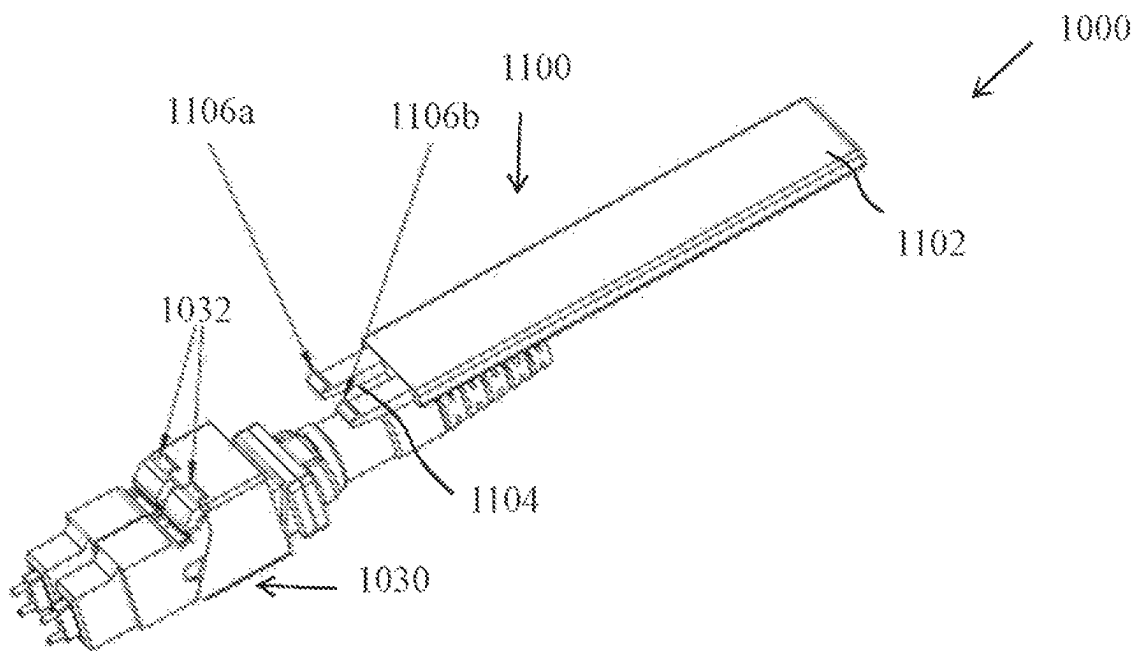


FIG. 11A

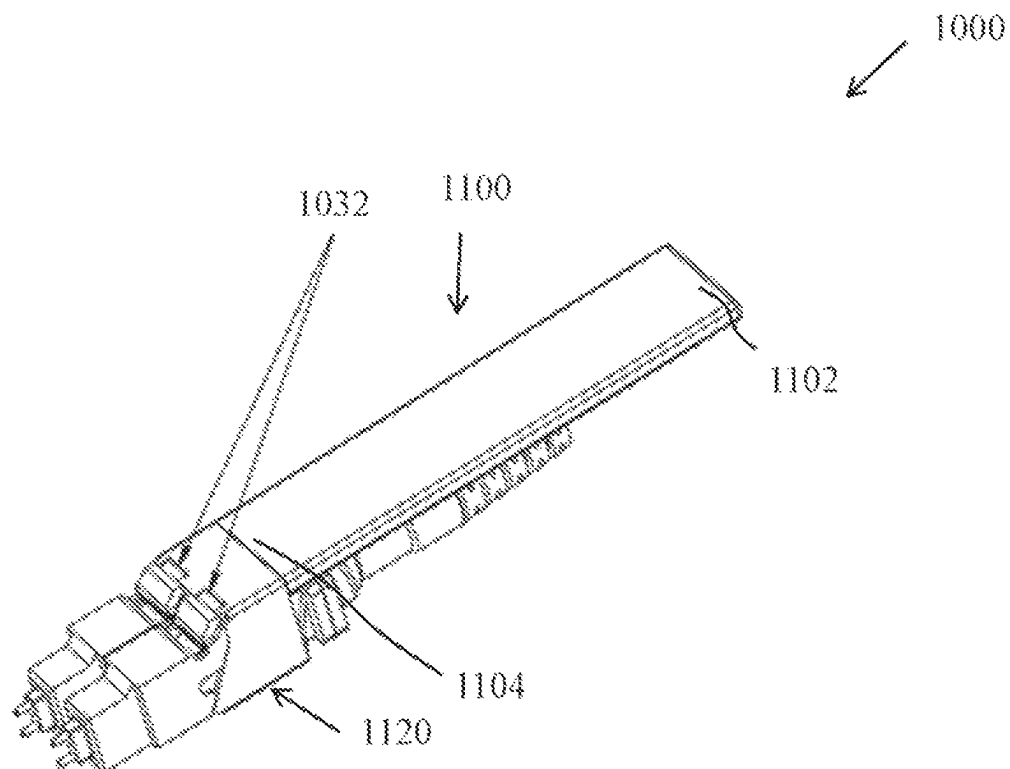


FIG. 11B

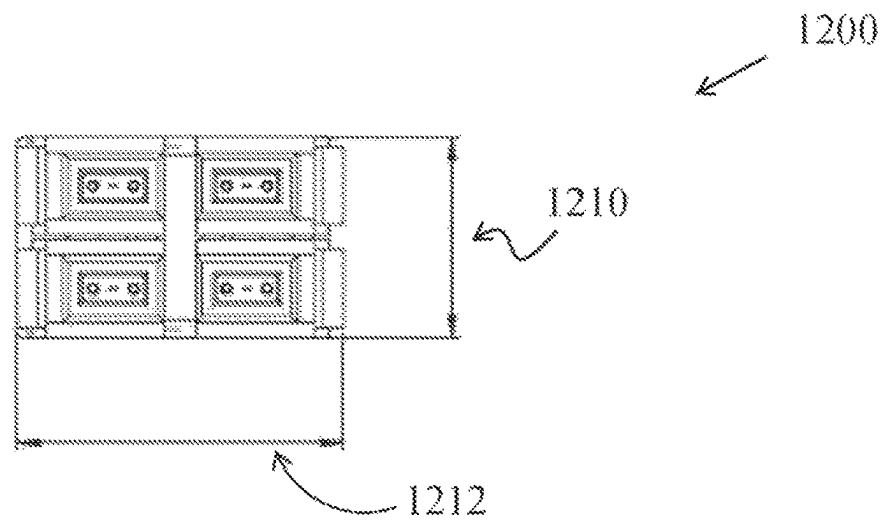


FIG. 12A

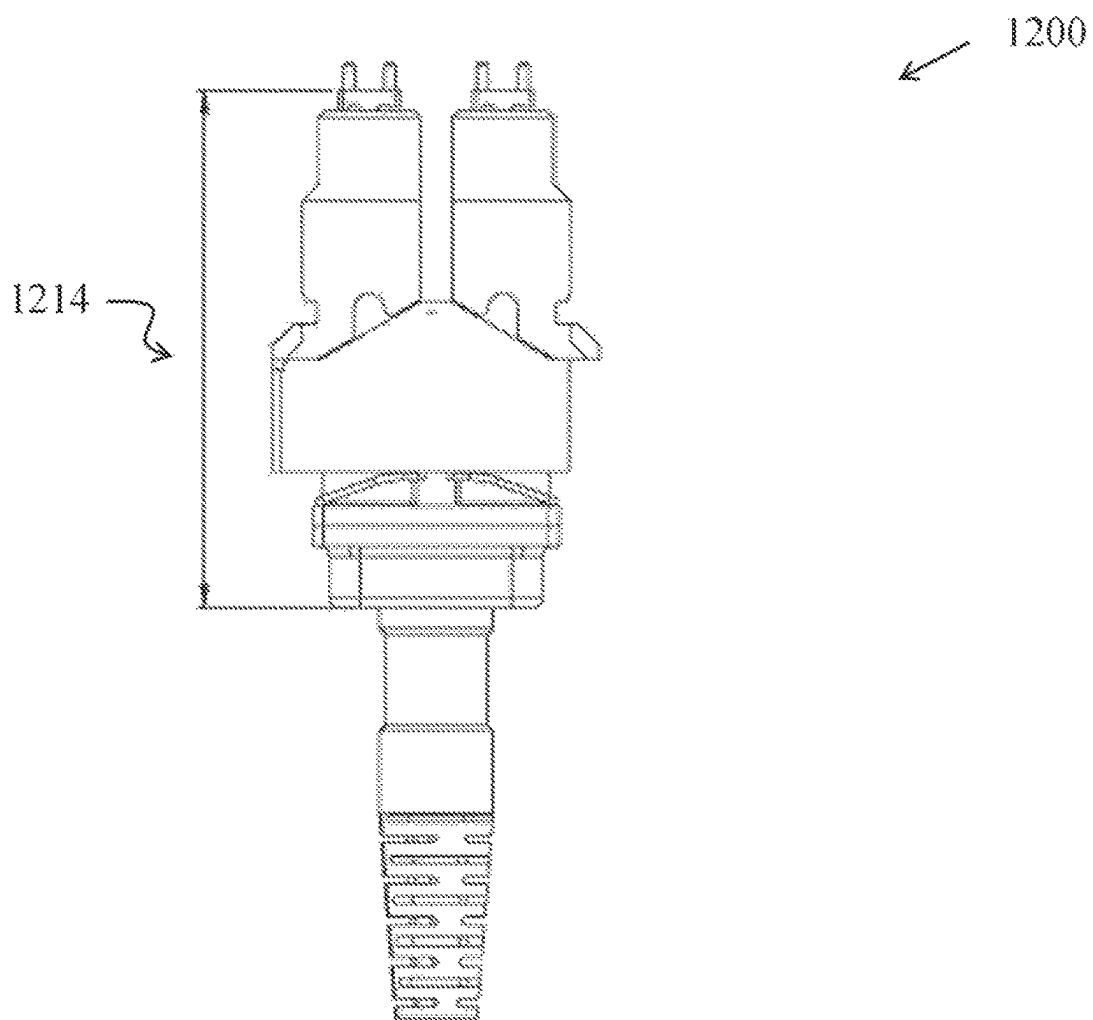
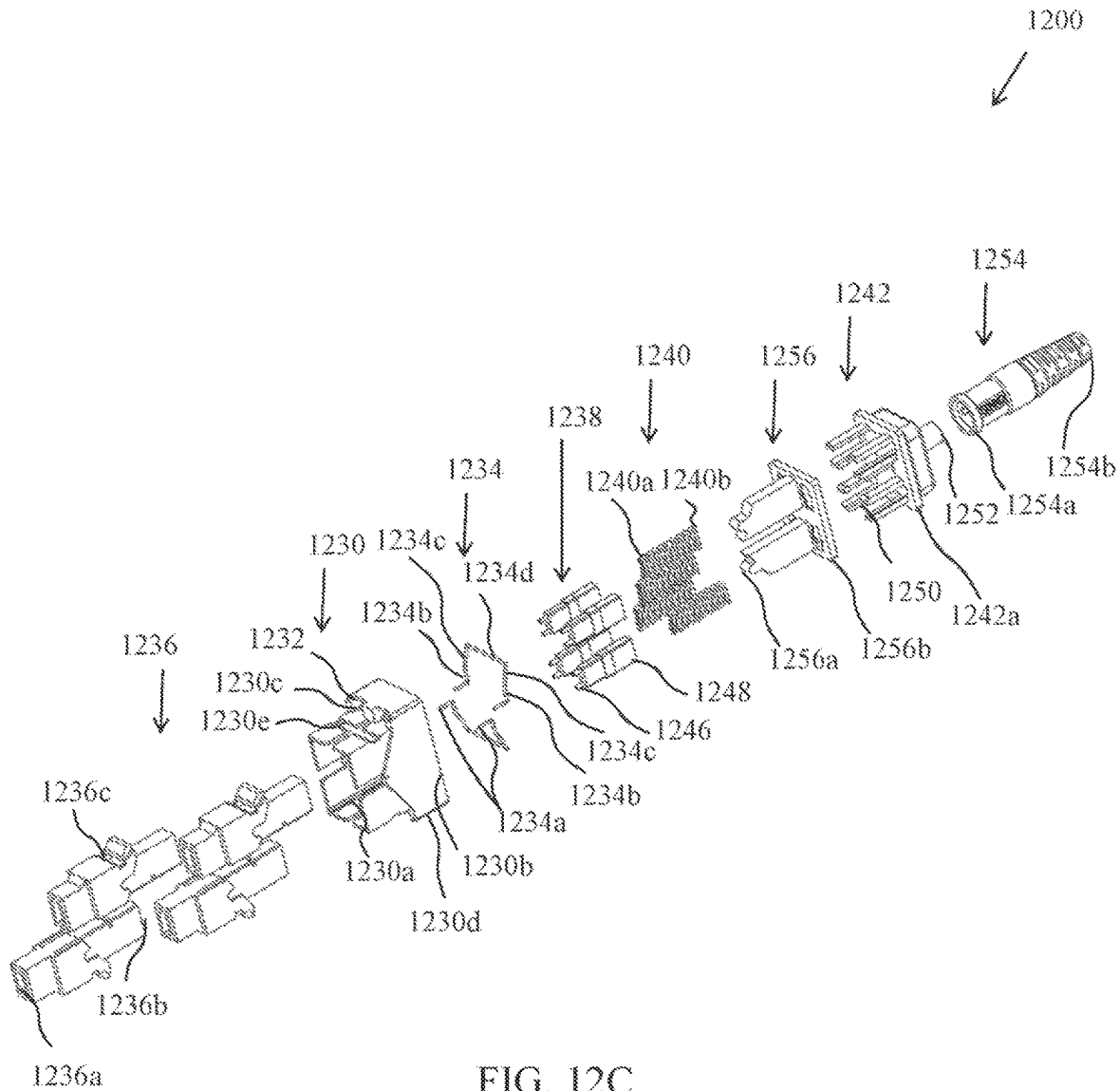


FIG. 12B



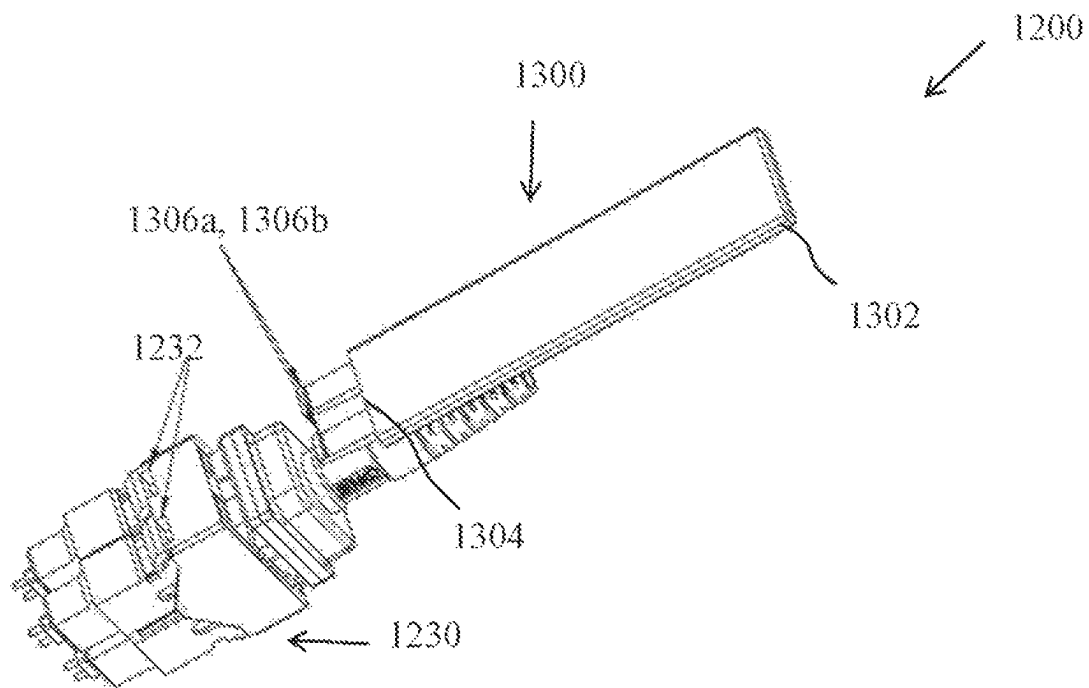


FIG. 13A

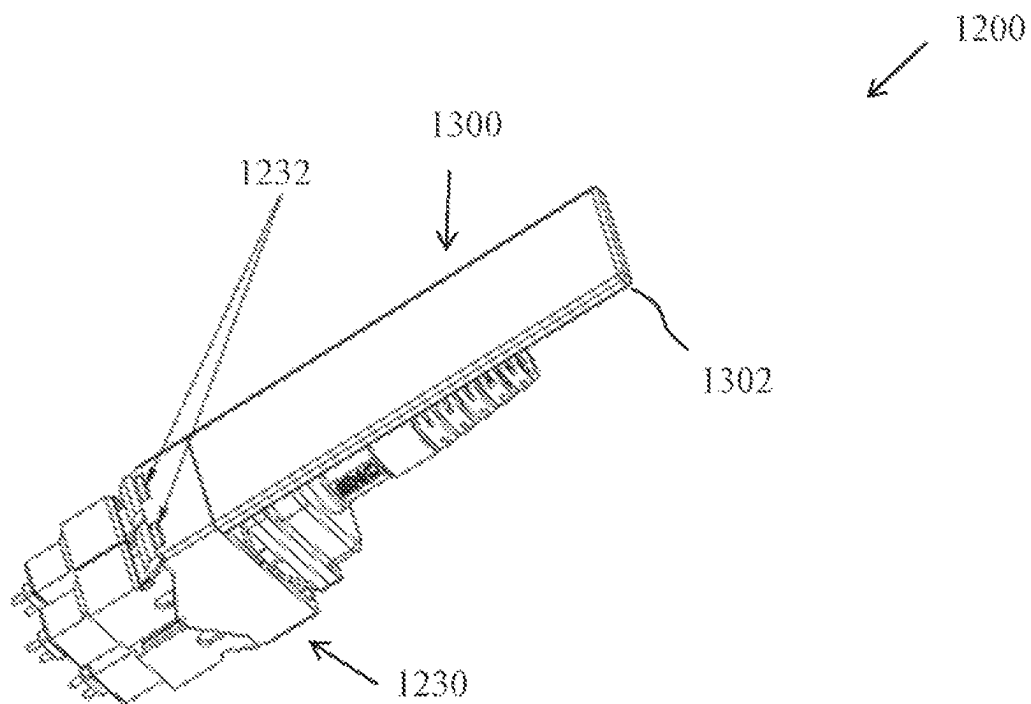


FIG. 13B

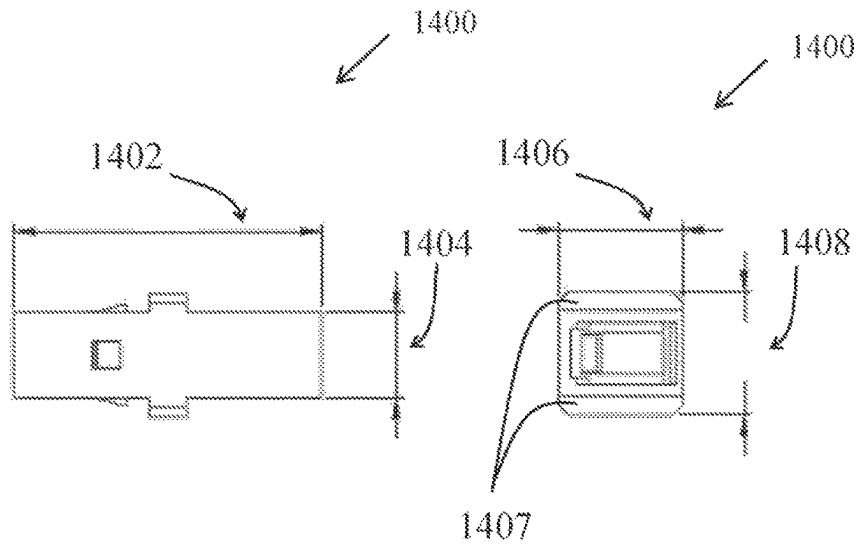


FIG. 14A

FIG. 14B

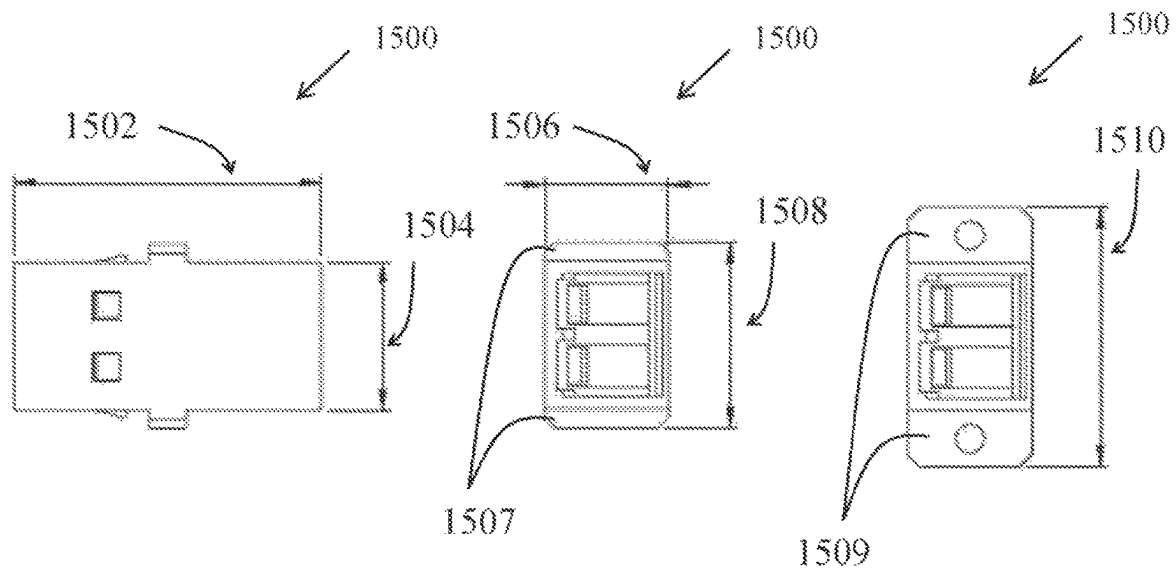


FIG. 15A

FIG. 15B

FIG. 15C

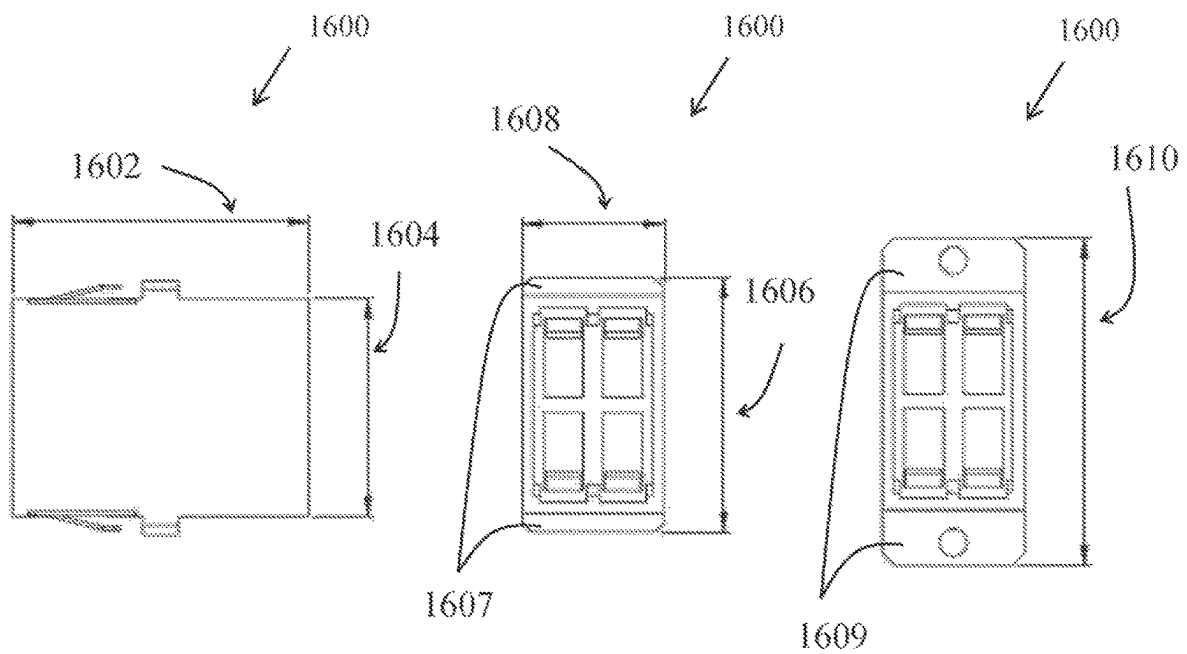


FIG. 16A

FIG. 16B

FIG. 16C

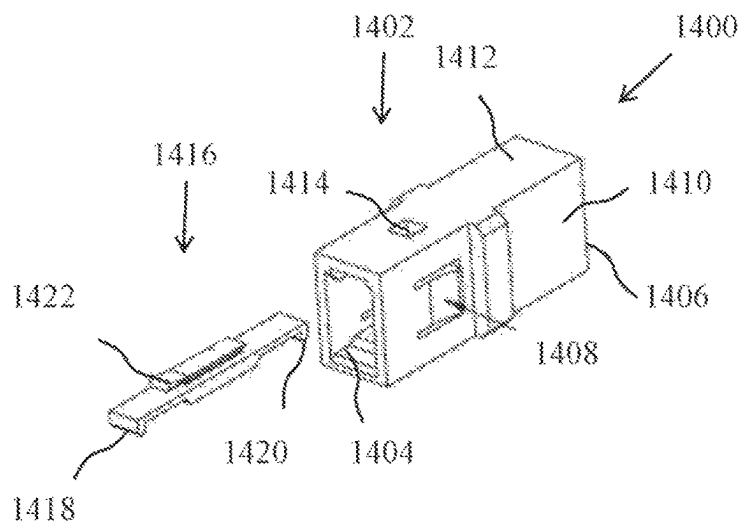


FIG. 17A

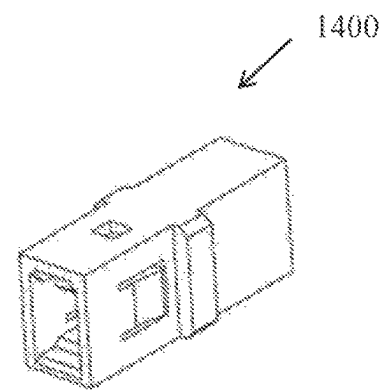


FIG. 17B

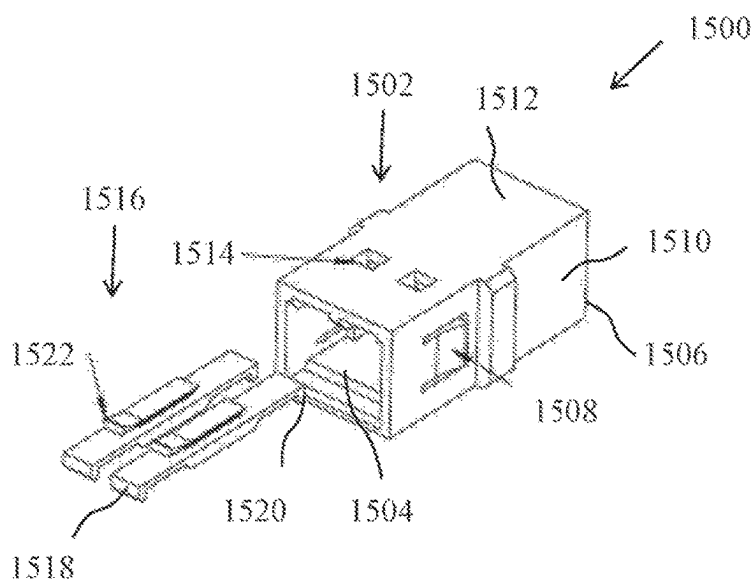


FIG. 18A

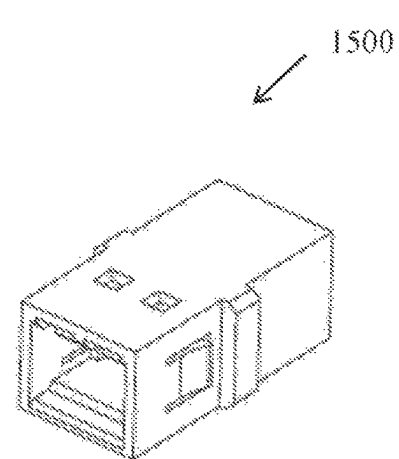
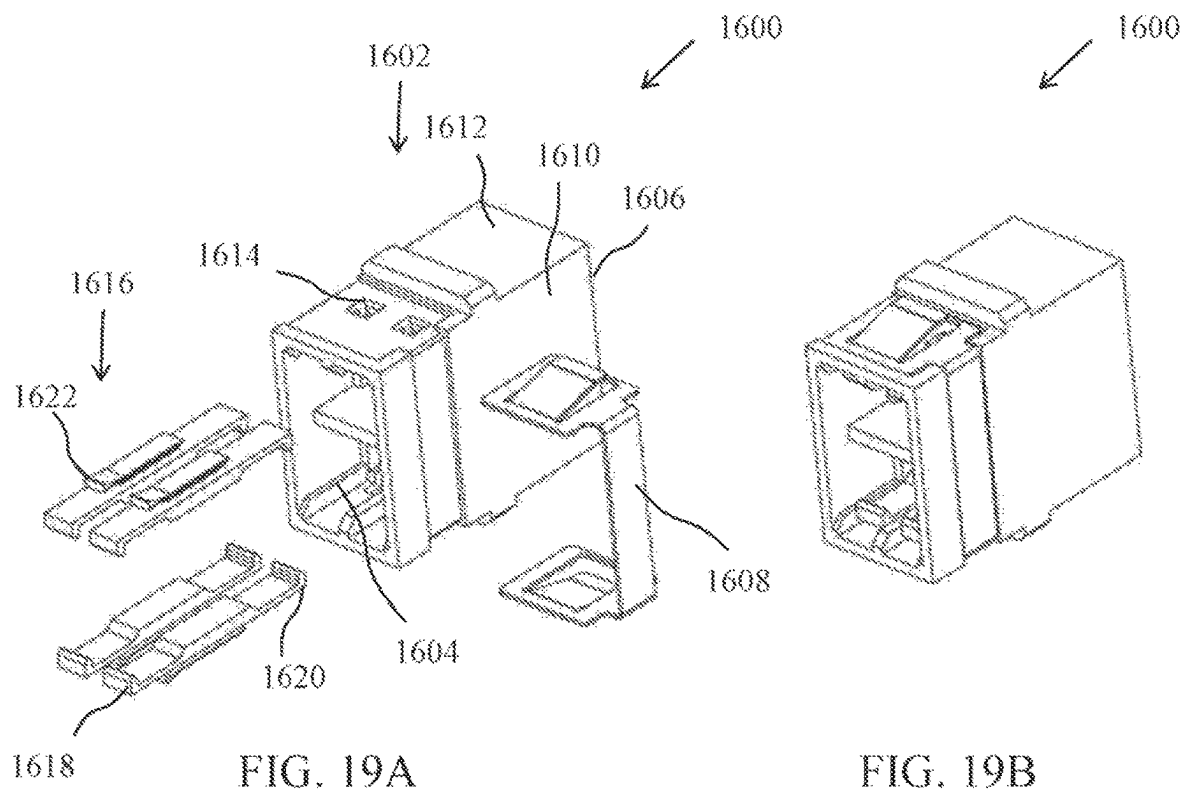


FIG. 18B



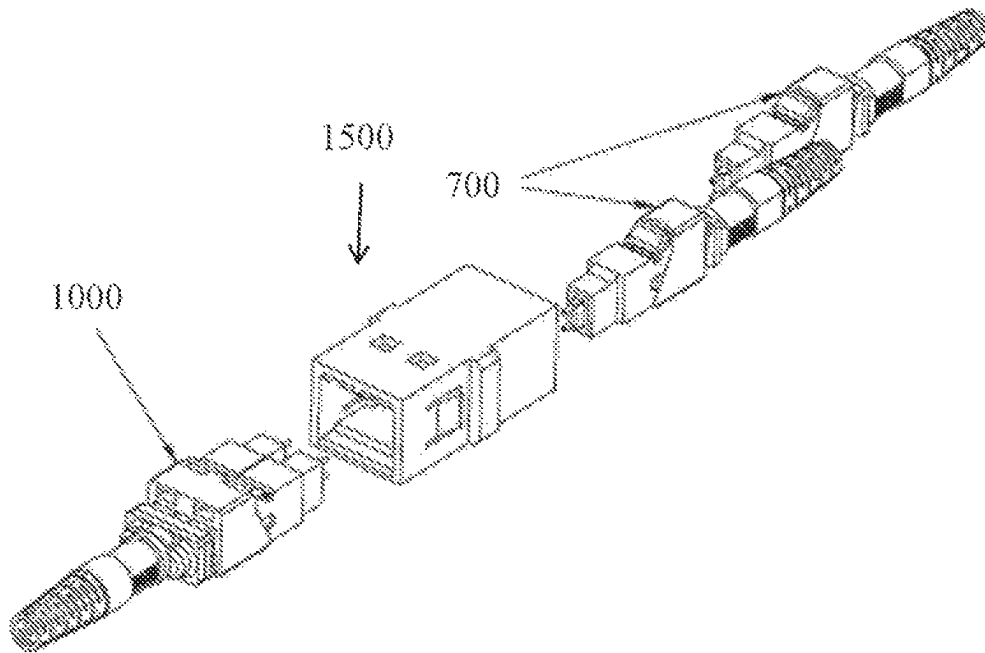


FIG. 20A

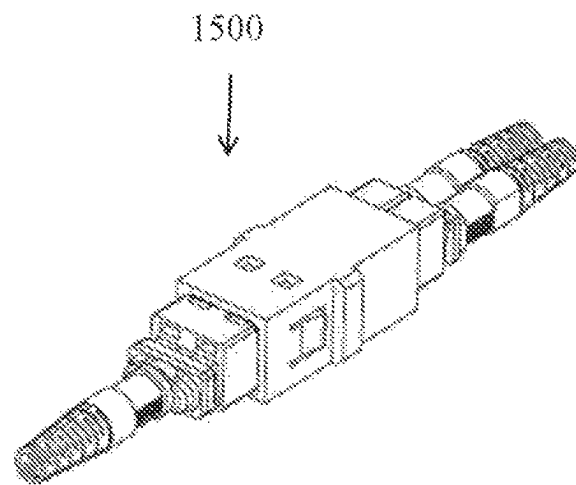


FIG. 20B

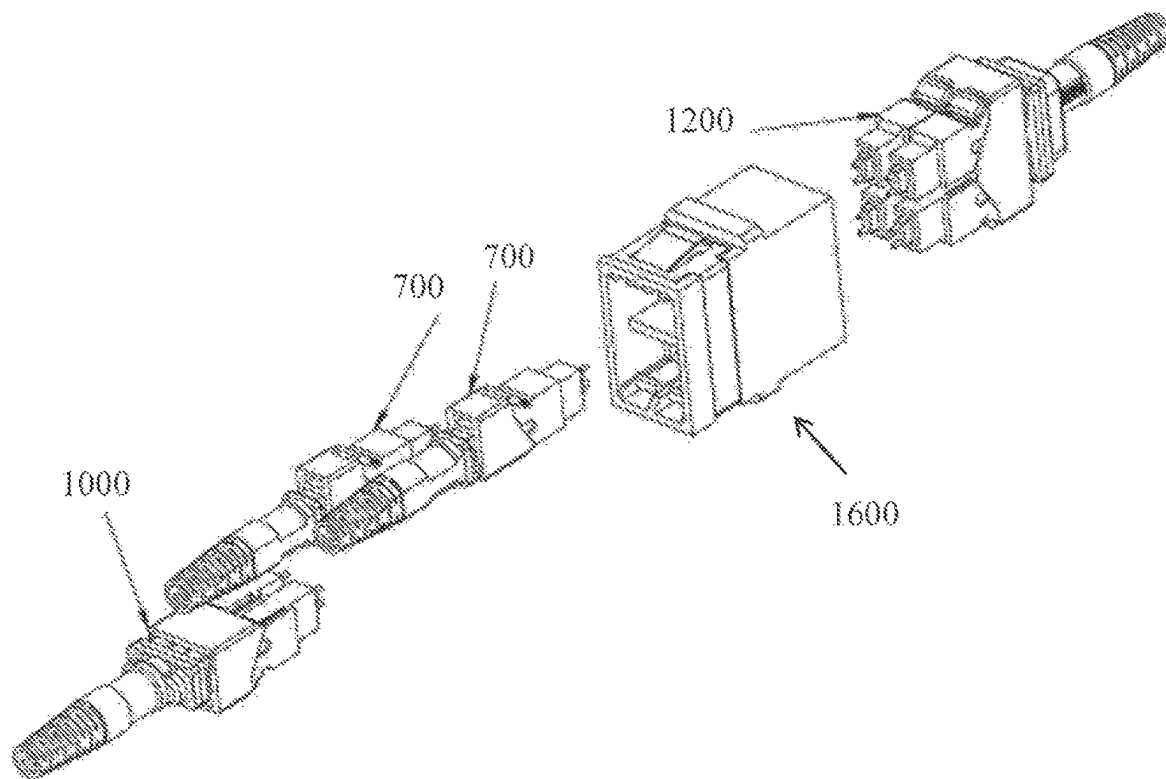


FIG. 21A

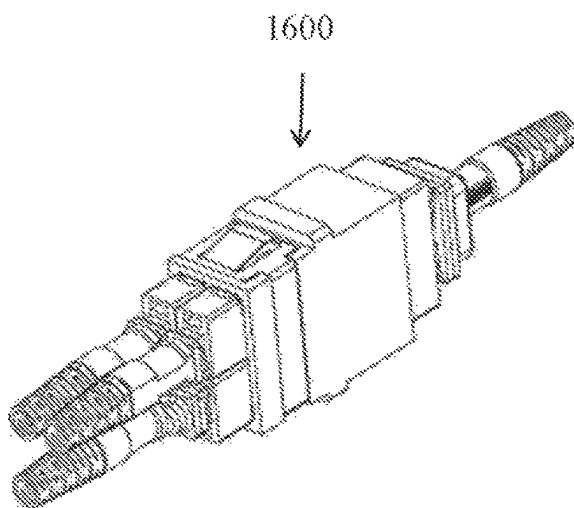


FIG. 21B

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ADAPTER SYSTEM FOR MULTI-FIBER MECHANICAL TRANSFER TYPE FERRULE

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. application Ser. No. 17/115,975 filed Dec. 9, 2020, which is a continuation of U.S. application Ser. No. 16/123,381, filed Sep. 6, 2018, now U.S. Pat. No. 10,866,371 issued on Dec. 15, 2020, which is a continuation of U.S. application Ser. No. 16/016,577 filed Jun. 23, 2018, now U.S. Pat. No. 10,107,972 issued on Oct. 23, 2018, a continuation of U.S. application Ser. No. 15/670,161 filed Aug. 7, 2017, now U.S. Pat. No. 10,078,187, issued on Sep. 18, 2018, which is a continuation of U.S. application Ser. No. 15/195,405 filed Jun. 28, 2016, now U.S. Pat. No. 9,726,830, issued on Aug. 8, 2017 and all are incorporated in their entireties.

FIELD OF THE INVENTION

The present teaching relates generally to connectors and adaptors for two-fiber mechanical transfer type ferrules, and more specifically to micro connectors and adaptors, such as simplex, duplex and quad micro connectors and adaptors with remote release and two-fiber ferrules.

BACKGROUND

The prevalence of the Internet has led to unprecedented growth in communication networks. Consumer demand for service and increased competition has caused network providers to continuously find ways to improve quality of service while reducing cost.

Certain solutions have included the deployment of high-density network panels. These panels are designed to consolidate the increasing volume of interconnections necessary to support the fast-growing networks into a compacted space, thereby increasing quality of service and decreasing costs such as floor space and support overhead. High-density network panels may contain a large number of ports to accommodate the large number of connections often necessary to connect the network panels with the network switches or other network end terminals. The panels' ports are generally designed to mate with connectors that contain two fibers (herein referred to as a 2-fiber connector) where one fiber transmits data (Tx) and the other fiber receives data (Rx). It is common to have a number of 2-fiber connectors, such as six 2-fiber connectors, from the network panel combined together into a fan-out jumper. In this design, the 2-fiber connectors and their fibers couple through a fan-out block and into a single cable, such as a 12-fiber cable. A multi-fiber push-on (MPO) connector, such as a 12-fiber MPO connector, is usually mounted at the end of the single cable to connect to a transceiver located at the network switch or other network end terminal. The transceiver can then manage a large number of fibers and 2-fiber connectors using the fan-out jumper and connect to a large number of ports from the network panels. However, the design, the connector dimensions and the connector accessibility of the presently available fan-out jumpers have not been fully optimized for high-density network panels.

Fan-out jumpers with 2-fiber Tx-Rx connectors may require individual ports for each Tx-Rx network connection; however, due to their design, they may create fiber management and maintenance issues for the operator that are challenging and undesirable. For instance, in a fan-out

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jumper with one 12-fiber MPO connector and six 2-fiber connectors, the MPO connector activates all 12 fibers as soon as it is plugged into the transceiver. However, this is convenient if there are six Tx-Rx ports that need service simultaneously on the network panel. This can cause some or all of the six Tx-Rx ports to become interrupted due to a malfunction with the MPO connector at the transceiver. Moreover, repairing the MPO connector will interrupt all 6 Tx-Rx ports as soon as the MPO connector is unplugged for troubleshooting, which may lead to an unnecessary interruption of services. Furthermore, if the network switch has 12-fiber MPO type transceivers and only one network panel Tx-Rx port coupled to a 2-fiber jumper connector needs service, this results in five additional 2-fiber connectors that are "active" but not utilized from the cable jumper. The unused but active 2-fiber connectors then need to be stored along with some lengths of 2-fiber cables until a future need is realized, adding to the fiber management problems. Additionally, 2-fiber connectors from one 12-fiber MPO fan-out jumper may not be able to reach Tx-Rx ports on the network panel needing service, in which case another transceiver would need to be powered up and another 12-fiber MPO fan-out jumper would need to be added to the equipment racks. If the length of the 12-fiber MPO fan-out jumper was determined based on the Tx-Rx port farthest from the transceiver, then there would be excessively long 2-fiber connector cables to manage when nearby Tx-Rx ports are serviced, again adding to the problem of fiber management. It is most desired to manage or "dress" the cables in horizontal and vertical cable troughs and not crossover the equipment in diagonal "short-cuts", as this causes equipment access and cable identification problems.

Certain available ferrules and connectors may be designed to couple with fan-out jumper and high-density network panels but their dimensions are not optimized for 2-fiber applications. For instance, MPO connectors are commonly configured for 2 to 36 fiber ribbon cables and are therefore much bigger than what is needed for two fiber applications with the same fiber-to-fiber center distance of about 0.25 mm. MPO connectors are about 7.6 mm high, 12.4 mm wide, 25.7 mm long (without including the strain relief boot), and MPO ferrules have a frontal contact surface of about 2.5 mm high and 6.4 mm wide. In contrast, MT-RJ connectors are configured for cable with a smaller number of fiber ribbons, such as 1 to 4 fibers, but they are larger than what is needed for two-fiber applications. MT-RJ connectors are about 10 mm high, 9.2 mm wide, 20 mm long (without including the strain relief boot), and MT-RJ ferrules have a frontal contact surface of about 2.5 mm high and 4.4 mm wide. Also, LC and SC duplex connectors may be designed for 2-fiber applications but their overall dimensions are not optimized to address the high-density requirement of high-density network panels. LC duplex connectors are about 12.7 mm high, 12.8 mm wide, 27.3 mm long (without including the strain relief boot), and SC duplex connectors are about 9.3 mm high, 24.2 mm wide, 25.2 mm long (without including the strain relief boot).

For 2-fiber jumper connectors coupled to fan-out jumpers and high-density network panels, adjacent connectors and cable assemblies may obstruct access to the individual release mechanisms at the Tx-Rx network panel port. Such physical obstructions may impede the ability of an operator to minimize the stresses applied to the cables and the connectors. For example, these stresses may be applied when the user reaches into a dense group of jumper connectors and pushes aside surrounding cables to access an individual connector release mechanism with his/her thumb

and forefinger. Overstressing the cables and connectors may produce latent defects, compromise the integrity and/or reliability of the terminations, and potentially cause serious disruptions to network performance.

While an operator may attempt to use a tool, such as a screwdriver, to reach into the dense group of 2-fiber jumper connectors and activate the release mechanism, the adjacent cables and connectors may obstruct the operator's line of sight, making it difficult to guide the tool to the release mechanism without pushing aside the surrounding cables. Moreover, even when the operator has a clear line of sight, guiding the tool to the release mechanism may be a time-consuming process. Thus, using a tool may not be effective at reducing support time and increasing the quality of service.

Accordingly, there remains a critical need for a better designed fan-out jumper that meets the requirements of high-density network panels.

SUMMARY

In one aspect, a connector is disclosed for two-fiber mechanical transfer type ferrule. The connector may comprise, for example and without limitation, at least one housing, a latch body, at least one latch spring, at least one ferrule, at least one ferrule spring, and a pull-tab. The connector can be, for example and without limitation, a micro-mechanical transfer (micro-MT) type connector, and more specifically a simplex micro-MT connector, a duplex micro-MT connector, or a quad micro-MT connector.

In some embodiments, the micro-MT connector contains at least one ferrule. The ferrule can be a micro-MT ferrule, and more specifically a male micro-MT ferrule or a female micro-MT ferrule. The male micro-MT ferrule and the female micro-MT ferrule have a contact surface with a height in a range of about 1.0 mm to about 2.4 mm, such as a height of about 2.0 mm, or a height of about 1.6 mm. The male micro-MT ferrule and the female micro-MT ferrule have a contact surface with a width in a range of about 2.4 mm to about 4.3 mm, such a width of about 4.0 mm or a width of about 3.6 mm.

In some embodiments, the micro-MT connector is a simplex micro-MT connector comprising a housing defining a longitudinal passage therethrough. The simplex micro-MT connector comprises a latch body defining a longitudinal passage therethrough and being movably coupled to the housing. The simplex micro-MT connector comprises a latch spring defining a longitudinal passage therethrough and being movably coupled with the housing. The simplex micro-MT connector comprises one ferrule extending longitudinally from a first end to a second end, the ferrule disposed at least partially within the longitudinal passage of the housing with the first end protruding external to the longitudinal passage. The simplex micro-MT connector comprises a ferrule spring dispersed in the longitudinal passage of the housing and coupled to the ferrule.

In some embodiments, the micro-MT connector is a simplex micro-MT connector having a height in a range of about 3.8 mm to about 7.5 mm, such a height of about 6.5 mm or a height of about 5.5 mm. The simplex micro-MT connector has a width in a range of about 6.4 mm to about 9.1 mm, such a width of about 8.8 mm or a width of about 8.5 mm. The simplex micro-MT connector has a length in a range of about 17.0 mm to about 25.1 mm, such a length of about 23.2 mm or a length of about 21.3 mm.

In some embodiments, the micro-MT connector is a simplex micro-MT connector comprising a pull-tab. The

pull-tab comprises a proximal end and a distal end spaced from the proximal end. The proximal end may include a latch hook uniquely shaped so as to engage with the unique profile of the latch body that is integral to the simplex micro-MT connector.

In some embodiments, the micro-MT connector is a simplex micro-MT connector comprising a latch body and a latch spring. When the simplex micro-MT connector is coupled to an adaptor, the latch spring is not compressed and therefore not actuated. A latch hole from the housing and a latch hole from the latch body are disposed vertically, one above each other, to generate a large cavity where a latch from the adaptor can couple to fasten the connector to the adaptor. In contrast, when the simplex micro-MT connector is decoupled from an adaptor, the latch spring is compressed and therefore actuated. The latch hole from the housing and the latch hole from the latch body are not disposed vertically, one above each other, and do not generate a large enough cavity where the latch from the adaptor can couple to fasten the connector to the adaptor.

In some embodiments, the micro-MT connector is a duplex micro-MT connector comprising a first housing and a second housing each defining a longitudinal passage therethrough from a proximal end to a distal end and including a latch hole designed to couple with a latch arm from an adaptor. The duplex micro-MT connector comprises a latch body defining a first longitudinal passage therethrough from a proximal end to a distal end spaced from the first end, a second longitudinal passage therethrough from a middle to the distal end spaced from the middle. The latch body is movably coupled with the housings wherein at least part of the housing is movably within at least the first longitudinal passage of the latch body. The duplex micro-MT connector comprises a latch spring defining a longitudinal passage therethrough and being movably coupled with the housing. The duplex micro-MT connector comprises a first ferrule and a second ferrule each extending longitudinally from a first end to a second end, the ferrule disposed at least partially within the longitudinal passage of the housing with the first end protruding external to the longitudinal passage. The duplex micro-MT connector comprises a first ferrule spring and a second ferrule spring each dispersed in the longitudinal passage of the housing and coupled to the ferrule.

In some embodiments, the micro-MT connector is a duplex micro-MT connector having a height in a range of about 7.8 mm to about 13.8 mm, such as a height of about 12.8 mm or a height of about 11.8 mm. The duplex micro-MT connector has a width in a range of about 6.4 mm to about 9.1 mm, such as a width of about 8.8 mm or a width of about 8.5 mm. The duplex micro-MT connector has a length in a range of about 19.4 mm to about 29.0 mm, such as a length of about 27.0 mm or a length of about 25.0 mm.

In some embodiments, the micro-MT connector is a duplex micro-MT connector comprising a pull-tab. The pull-tab comprises a proximal end and a distal end spaced from the proximal end. The proximal end may include a first latch hook and a second latch hook uniquely shaped so as to engage with the unique profile of the latch body that is integral to the duplex micro-MT connector.

In some embodiments, the micro-MT connector is a quad micro-MT connector comprising a first housing and a second housing and a third housing and a fourth housing each defining a longitudinal passage therethrough from a proximal end to a distal end and including a latch hole designed to couple with a latch arm from an adaptor. The quad micro-MT connector comprises a latch body defining a first

longitudinal passage therethrough from a proximal end to a distal end, a second longitudinal passage therethrough from a middle to the distal end. The latch body is movably coupled with the housings wherein at least part of the housing is movably within at least the first longitudinal passage of the latch body. The quad micro-MT connector comprises a first latch spring and a second latch spring each defining a longitudinal passage therethrough and being movably coupled with the housing. The quad micro-MT connector comprises a first ferrule and a second ferrule and a third ferrule and a fourth ferrule each extending longitudinally from a first end to a second end, the ferrule disposed at least partially within the longitudinal passage of the housing with the first end protruding external to the longitudinal passage. The quad micro-MT connector comprises a first ferrule spring and second ferrule spring and a third ferrule spring and a fourth ferrule spring each dispersed in the longitudinal passage of the housing and coupled to the ferrule.

In some embodiments, the micro-MT connector is a quad micro-MT connector having a height in a range of about 7.8 mm to about 13.8 mm, such as a height of about 12.8 mm or a height of about 11.8 mm. The quad micro-MT connector has a width in a range of about 12.8 mm to about 21.2 mm, such as a width of about 20.2 mm or a width of about 19.2 mm. The quad micro-MT connector has a length in a range of about 19.4 mm to about 29.0 mm, such as a length of about 27.0 mm or a length of about 25.0 mm.

In some embodiments, the micro-MT connector is a quad micro-MT connector comprising a pull-tab. The pull tab comprises a proximal end and a distal end spaced from the proximal end. The proximal end may include a first latch hook and a second latch hook uniquely shaped so as to engage with the unique profile of the latch body that is integral to the quad micro-MT connector.

In another aspect, an adaptor is disclosed for two-fiber mechanical transfer type ferrule. The adaptor may comprise, for example and without limitation, a housing and at least one latch hook arm. The adaptor can be, for example and without limitation, a micro-mechanical transfer (micro-MT) type adaptor, and more specifically a simplex micro-MT adaptor, a duplex micro-MT adaptor, or a quad micro-MT adaptor.

In some embodiments, the micro-MT adaptor is a simplex micro-MT adaptor comprising a housing defining a longitudinal passage therethrough, the housing comprising at least one mounting tab, wherein a first wall, a second wall, a third wall, and a fourth wall, wherein the first wall and the third wall are opposite to each other, wherein the second wall and the fourth wall are opposite to each other, and wherein the second wall comprises at least one adaptor housing hole. The simplex micro-MT adaptor comprises a latch-hook arm defined by a first side, a second side, a first end, and a second end spaced from the first end, wherein the first end and the second end each comprise a first hook configured to couple with the connector, wherein one of the sides comprises a second hook to couple with an adaptor housing hole.

In some embodiments, the micro-MT adaptor is a simplex micro-MT adaptor, which may have a length in a range of about 22 mm to about 28 mm, such as a length of about 27 mm or a length of about 26 mm. The simplex micro-MT adaptor may have a height in a range of about 4 mm to about 10 mm, such as a height of about 9 mm or a height of about 8 mm. The simplex micro-MT adaptor may have a width in a range of about 7.2 mm to about 13.2 mm, such as a width of about 12.2 mm or a width of about 11.2 mm. The simplex

micro-MT adaptor may have mounting handles and a height in a range of about 7 mm to about 13 mm, such as a height of about 12 mm or a height of about 11 mm.

In some embodiments, the micro-MT adaptor is a duplex micro-MT adaptor comprising a housing defining a longitudinal passage therethrough, the housing comprising at least one mounting tab, wherein a first wall, a second wall, a third wall, and a fourth wall, wherein the first wall and the third wall are opposite to each other, wherein the second wall and the fourth wall are opposite to each other, and wherein the second wall comprises at least one adaptor housing hole. The duplex micro-MT adaptor comprises two latch-hook arms each defined by a first side, a second side, a first end, and a second end spaced from the first end, wherein the first end and the second end each comprise a first hook configured to couple with the connector, wherein one of the sides comprises a second hook to couple with an adaptor housing hole.

In some embodiments, the micro-MT adaptor is a duplex micro-MT adaptor, which may have a length in a range of about 22 mm to about 28 mm, such as a length of about 27 mm or a length of about 26 mm. The duplex micro-MT adaptor may have a height in a range of about 9 mm to about 15 mm, such as a height of about 14 mm or a height of about 13 mm. The duplex micro-MT adaptor may have a width in a range of about 7.2 mm to about 13.2 mm, such as a width of about 12.2 mm or a width of about 11.2 mm. The duplex micro-MT adaptor may have mounting handles and a height in a range of about 12 mm to about 18 mm, such as a height of about 17 mm or a height of about 16 mm. The duplex micro-MT adaptor may have flanges and a height in a range of about 18 mm to about 24 mm, such as a height of about 23 mm or a height of about 22 mm.

In some embodiments, the micro-MT adaptor is a quad micro-MT adaptor comprising a housing defining a longitudinal passage therethrough, the housing comprising at least one mounting tab, wherein a first wall, a second wall, a third wall, and a fourth wall, wherein the first wall and the third wall are opposite to each other, wherein the second wall and the fourth wall are opposite to each other, and wherein the second wall comprises at least one adaptor housing hole. The quad micro-MT adaptor comprises four latch-hook arms each defined by a first side, a second side, a first end, and a second end spaced from the first end, wherein the first end and the second end each comprise a first hook configured to couple with the connector, wherein one of the sides comprises a second hook to couple with an adaptor housing hole.

In some embodiments, the micro-MT adaptor is a quad micro-MT adaptor, which may have a length in a range of about 22 mm to about 28 mm, such as a length of about 27 mm or a length of about 26 mm. The quad micro-MT adaptor may have a height in a range of about 15.3 mm to about 21.3 mm, such as a height of about 20.3 mm or a height of about 19.3 mm. The duplex micro-MT adaptor may have a width in a range of about 9 mm to about 15 mm, such as a width of about 14 mm or a width of about 13 mm. The duplex micro-MT adaptor may have mounting handles and a height in a range of about 18.3 mm to about 24.3 mm, such as a height of about 23.3 mm or a height of about 22.3 mm. The duplex micro-MT adaptor may have flanges and a height in a range of about 24.5 mm to about 30.5 mm, such as a height of about 29.5 mm or a height of about 28.5 mm.

In another aspect, a micro-MT adaptor is used to couple at least one micro-MT connector. The connector can be, for example and without limitation, a micro-MT type connector, and more specifically a simplex micro-MT connector, a

duplex micro-MT connector, or a quad micro-MT connector. The adaptor can be, for example and without limitation, a micro-MT type adaptor, and more specifically a simplex micro-MT adaptor, a duplex micro-MT adaptor, or a quad micro-MT adaptor.

In some embodiments, the micro-MT adaptor is a duplex micro-MT adaptor coupling with a duplex micro-MT connector, a first simplex micro-MT connector, and a second simplex micro-MT connector.

In some embodiments, the micro-MT adaptor is a quad micro-MT adaptor coupling with a first simplex micro-MT connector, a second simplex micro-MT connector, a duplex micro-MT connector, and a quad micro-MT connector.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1A is a perspective view of a prior art standard MPO connector;

FIG. 1B is a perspective view of a prior art standard MPO male ferrule;

FIG. 1C is a perspective view of a prior art standard MPO female ferrule;

FIG. 1D is a front view of a prior art standard MPO connector;

FIG. 1E is a top view of a prior art standard MPO connector;

FIG. 2A is a perspective view of a prior art standard MT-RJ connector;

FIG. 2B is a perspective view of a prior art standard MT-RJ male ferrule;

FIG. 2C is a perspective view of a prior art standard MT-RJ female ferrule;

FIG. 2D is a front view of a prior art standard MT-RJ connector;

FIG. 2E is a top view of a prior art standard MT-RJ connector;

FIG. 3A is a front view of a prior art standard LC duplex connector;

FIG. 3B is a top view of a prior art standard LC duplex connector;

FIG. 4A is a front view of a prior art standard SC duplex connector;

FIG. 4B is a top view of a prior art standard SC duplex connector;

FIG. 5A is a front view of a prior art standard 12-fiber MPO connector;

FIG. 5B is a schematic representation of a prior art standard LC duplex fan-out jumper, showing six 2-fiber connectors;

FIG. 6 is a schematic representation of network switch and distribution equipment racks, showing the MPO connector end of a jumper at a transceiver;

FIG. 7A is a perspective view of one embodiment of a simplex micro-MT connector according to aspects of the present disclosure;

FIG. 7B is a perspective view of one embodiment of a micro-MT male ferrule according to aspects of the present disclosure;

FIG. 7C is a perspective view of one embodiment of a micro-MT female ferrule according to aspects of the present disclosure;

FIG. 7D is a front view of one embodiment of a simplex micro-MT connector according to aspects of the present disclosure;

FIG. 7E is a top view of one embodiment of a simplex micro-MT connector according to aspects of the present disclosure;

FIG. 7F is an exploded view of one embodiment of a simplex micro-MT connector according to aspects of the present disclosure;

FIG. 8A is a perspective view of one embodiment of a simplex micro-MT connector according to aspects of the present disclosure, showing a pull-tab;

FIG. 8B is a perspective view of one embodiment of a simplex micro-MT connector according to aspects of the present disclosure, showing the pull-tab of FIG. 8A engaged in a hook hole;

FIG. 9A is a perspective view of one embodiment of a simplex micro-MT connector according to aspects of the present disclosure, showing a relaxed latch spring;

FIG. 9B is a perspective view of one embodiment of a simplex micro-MT connector according to aspects of the present disclosure, showing a compressed latch spring;

FIG. 10A is a front view of one embodiment of a duplex micro-MT connector according to aspects of the present disclosure;

FIG. 10B is a top view of one embodiment of a duplex micro-MT connector according to aspects of the present disclosure;

FIG. 10C is an exploded view of one embodiment of a duplex micro-MT connector according to aspects of the present disclosure;

FIG. 11A is a perspective view of one embodiment of a duplex micro-MT connector according to aspects of the present disclosure, showing a pull-tab;

FIG. 11B is a perspective view of one embodiment of a duplex micro-MT connector according to aspects of the present disclosure, showing the pull-tab of FIG. 11A engaged in a hook hole;

FIG. 12A is a front view of one embodiment of a quad micro-MT connector according to aspects of the present disclosure;

FIG. 12B is a top view of one embodiment of a quad micro-MT connector according to aspects of the present disclosure;

FIG. 12C is an exploded view of one embodiment of a quad micro-MT connector according to aspects of the present disclosure;

FIG. 13A is a perspective view of one embodiment of a quad micro-MT connector according to aspects of the present disclosure, showing a pull-tab;

FIG. 13B is a perspective view of one embodiment of a quad micro-MT connector according to aspects of the present disclosure, showing the pull-tab of FIG. 13A engaged in a latch hook;

FIG. 14A is a top view of one embodiment of a simplex micro-MT adapter according to aspects of the present disclosure;

FIG. 14B is an end view of one embodiment of a simplex micro-MT adapter without flanges according to aspects of the present disclosure;

FIG. 15A is a top view of one embodiment of a duplex micro-MT adapter according to aspects of the present disclosure;

FIG. 15B is an end view of one embodiment of a duplex micro-MT adapter without flanges according to aspects of the present disclosure;

FIG. 15C is an end view of one embodiment of a duplex micro-MT adapter with flanges according to aspects of the present disclosure;

FIG. 16A is a top view of one embodiment of a quad micro-MT adapter according to aspects of the present disclosure;

FIG. 16B is an end view of one embodiment of a quad micro-MT adapter without flanges according to aspects of the present disclosure;

FIG. 16C is an end view of one embodiment of a quad micro-MT adapter with flanges according to aspects of the present disclosure;

FIG. 17A is an exploded view of one embodiment of a simplex micro-MT adapter according to aspects of the present disclosure;

FIG. 17B is a perspective view of one embodiment of a simplex micro-MT adapter according to aspects of the present disclosure;

FIG. 18A is an exploded view of one embodiment of a duplex micro-MT adapter according to aspects of the present disclosure;

FIG. 18B is a perspective view of one embodiment of a duplex micro-MT adapter according to aspects of the present disclosure;

FIG. 19A is an exploded view of one embodiment of a quad micro-MT according to aspects of the present disclosure;

FIG. 19B is a perspective view of one embodiment of a quad micro-MT adapter according to aspects of the present disclosure;

FIG. 20A is an exploded view of one embodiment of a duplex micro-MT connector coupled to a first simplex micro-MT connector and a second simplex micro-MT connector through a duplex micro-MT adapter according to aspects of the present disclosure;

FIG. 20B is a perspective view of the duplex micro-MT connector coupled to a first simplex micro-MT connector and a second simplex micro-MT connector through the duplex micro-MT adapter of FIG. 20A;

FIG. 21A is an exploded view of one embodiment of a duplex micro-MT connector, a first simplex micro-MT connector, and a second simplex micro-MT connector coupled to a quad micro-MT connector through a quad micro-MT adapter according to aspects of the present disclosure; and

FIG. 21B is a perspective view of the duplex micro-MT connector, the first simplex micro-MT connector, and the second simplex micro-MT connector coupled to a quad micro-MT connector through the quad micro-MT adapter of FIG. 21A.

DETAILED DESCRIPTION

Certain exemplary embodiments will now be described to provide an overall understanding of the principles of the structure, function, and use of the devices and methods disclosed herein. One or more examples of these embodiments are illustrated in accompanying drawings. Those skilled in the art will understand that the devices and methods specifically described herein and illustrated in the accompanying drawings are non-limiting exemplary embodiments and that the scope of the present invention is defined solely by the claims. The features illustrated or described in connection with one exemplary embodiment may be combined with the features of other embodiments. Such embodiments and variations are intended to be included within the scope of the present invention.

So that the present invention may more readily be understood, certain terms are first defined.

As used herein, the terms “about” or “approximately” or “substantially” for any numerical values or ranges indicate a suitable dimensional tolerance that allows the part or

collection of elements to function for its intended purpose as described herein. These terms indicate a $\pm 10\%$ variation about a central value.

The term “ferrule” is used herein consistent with its customary meaning in the art to refer to a device and/or components thereof that connect optical fibers.

The term “connector” is used herein consistent with its customary meaning in the art to refer to a device and/or components thereof that connects a first module or cable to a second module or cable. The connectors may be configured for fiber optic transmission or electrical signal transmission. The connector may be any suitable type now known or later developed such as, for example and without any limitation, a ferrule connector (FC), a fiber distributed data interface (FDDI) connector, a LC connector, a mechanical transfer (MT) connector, a mechanical transfer registered jack (MT-RJ) connector, a SC connector, a SC duplex connector, a straight tip (ST) connector, a multiple-fiber push-on (MPO) connector, or a media interface connector (MIC).

The term “adaptor” is used herein consistent with its customary meaning in the art to refer to a device and/or components thereof that couples a first module or cable or set of cables with a second module or cable or set of cables.

The term “cable” is used herein consistent with its customary meaning in the art to refer to an insulated wire or wires comprising a protective casing (e.g., a jacket, a sheathing material) and used for transmitting electricity or telecommunication signals. A “fiber optic cable” or an “optical cable” referred herein to a cable containing one or more optical fibers for conducting optical signals in beams of light. The optical fibers can be constructed from any suitable transparent material, including glass, fiberglass, and plastic. In addition, the cable can be connected to a connector on one end or on both ends of the cable, the cable can be coupled to an adaptor on one end or on both ends of the cable, and the cable can be connected to another cable on one end or on both ends of the cable.

The term “jumper” is used herein consistent with its customary meaning in the art to refer to a device and/or components thereof that connect multicore cables.

The term “panel” is used herein consistent with its customary meaning in the art to refer to a device and/or components thereof that connect incoming and outgoing lines of communication cables.

The present application relates generally to devices and methods for connecting optical fibers. For example and without any limitation, the various devices and methods of the invention can be utilized to optimize the connections between ports in high-density network panels by increasing the accessibility to the connectors by the operator and by reducing the number of connector cables, the length of the connector cables, the number of connector cables that are dressed crossover to the equipment, and the dimension of the connectors. As discussed in more detail below, in some embodiments, optical fiber cables can be modularly coupled with one or more optical fiber connectors, adaptors, and other optical fiber cables using a remote release. Various aspects of the prior art and invention are described below.

Those in the art will understand that the devices and methods described herein are more suitable at least for 2-fiber cable and high-density network panel applications than the devices and methods described in the prior art.

For instance, as shown in FIG. 1A, a conventional MPO connector **100** is commonly configured for 2 to 36 fiber ribbon cables and is therefore much bigger than what is needed for 2-fiber applications with the same fiber-to-fiber

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center distance of about 0.25 mm. The conventional MPO Connector **100** is mounted with a male MPO ferrule **102** as shown in FIG. **1B** or a female MPO ferrule **104** as shown in FIG. **1C**, and the MPO ferrules have a frontal contact surface height **106** of about 2.5 mm and a frontal contact surface width **108** of about 6.4 mm. FIG. **1D** shows a front view of the conventional MPO connector **100** with a height **110** of about 7.6 mm, and FIG. **1E** shows a side view of the conventional MPO connector **100** with a width **112** of about 12.4 mm and a length **114** of about 25.7 mm.

In another instance, as shown in FIG. **2A**, a conventional MT-RJ connector **200** is configured for cable with a smaller number of fiber ribbons, such as 1 to 4 fibers, but it is larger than what is needed for 2-fiber applications. The MT-RJ connector **200** is mounted with a male MT-RJ ferrule **202** as shown in FIG. **2B** or a female MT-RJ ferrule **204** as shown in FIG. **2C**, and the MT-RJ ferrules have a frontal contact surface height **206** of about 2.5 mm and a frontal contact surface width **208** of about 4.4 mm. FIG. **2D** shows a front view of the conventional MT-RJ connector **200** with a height **210** of about 10 mm and a width **212** of about 9.2 mm, and FIG. **2E** shows a side view of the conventional MT-RJ connector **200** with a length **214** of about 20 mm.

In another instance, as shown in FIG. **3A**, a conventional LC duplex connector **300** is designed for 2-fiber applications but its overall dimension is not optimized to address the requirements of high-density network panels. The LC duplex connector **300** has a height **302** of about 12.7 mm and a width **304** of about 12.8 mm. FIG. **3B** shows a side view of the conventional LC duplex connector **300** with a length **306** of about 27.3 mm.

In another instance, as shown in FIG. **4A**, a conventional SC duplex connector **400** is designed for 2-fiber applications, similar to the LC duplex connector **300**, but its overall dimension is not optimized to address the requirements of high-density network panels. The SC duplex connector **400** has a height **402** of about 9.3 mm. FIG. **4B** shows a side view of the conventional SC duplex connector **400** with a width **404** of about 24.2 mm and a length **406** of about 25.2 mm.

In another example, FIG. **5B** shows a plan view of a conventional 12-fiber MPO—6 LC duplex fan-out jumper **502** that is often used to couple with a high-density network panel, and FIG. **5A** shows a front view of a 12-fiber MPO connector **506** that is attached at one end of the 12-fiber MPO—6 LC duplex fan-out jumper **502**. The 12-fiber MPO connector **506** comprises six fiber pairs **520**, **522**, **524**, **526**, **528**, and **530**, wherein each pair has one data-transmitting Tx fiber and one data-receiving Rx fiber. The twelve fibers go through a cable **508** and are separated into 2-fiber cables **514** at a fan-out block **512**. The 2-fiber cables **514** have a length **516** and are terminated by an LC connector **518** comprising one of the fiber pairs **520**, **522**, **524**, **526**, **528**, and **530**.

FIG. **6** shows a perspective view of the 12-fiber MPO—6 LC duplex fan-out jumper **500** and conventional network equipment racks **600** that can be found, for example, in a central office and data center. The network equipment racks **600** comprise a plurality of network switches **602** comprising a transceiver **604** and a plurality of network panels **606** comprising a plurality of network panel ports **608**. The network switches **602** and the network panels **604** are connected through the 12-fiber MPO—6 LC duplex jumper **500**. The 12-fiber MPO connector **506** (not shown) located at one end of the 12-fiber MPO—6 LC duplex fan-out jumper **500** connects to the transceiver **604**, and only four of the six 2-fiber LC cables **514** located at the other end of the 12-fiber MPO—6 LC duplex fan-out jumper **500** connect to

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the network panel ports **608**. The remaining two 2-fiber LC cables **514** do not connect to the network panel ports since the desired panel ports **608a** are not accessible due to the too short length **516** of the 2-fiber cables **514**. Therefore, the 12-fiber MPO—6 LC duplex fan-out jumper **500** is not optimized for a high-density network panel having network panel ports located at different distances from the 12-fiber MPO—6 LC duplex fan-out jumper **500**.

Various embodiments disclosed herein are configured for use as a mechanical transfer (MT) connector, such as a micro-MT (MMT) connector **700** as shown in FIG. **7A**, comprising a male micro-MT ferrule **702** as shown in FIG. **7B** or a female micro-MT ferrule **704** as shown in FIG. **7C**. The male micro-MT ferrule **702** and the female micro-MT ferrule **704** have a frontal contact surface smaller than that of conventional 2.5 mm high and 7.6 mm wide MPO ferrules and conventional 2.5 mm high and 4.4 mm wide MT-RJ ferrules. In some embodiments, the MT ferrule is a micro-MT ferrule having a contact surface with a height **706** in a range of about 1.0 mm to about 2.4 mm, such as a height of about 2.0 mm or a height of about 1.6 mm. In some embodiments, the MT ferrule is a micro-MT ferrule having a contact surface with a width **708** in a range of about 2.4 mm to about 4.3 mm, such as a width of about 4.0 mm or a width of about 3.6 mm.

FIG. **7D** shows a front view of a simplex micro-MT connector **700** having a height **710** smaller than that of a conventional 7.6 mm high MPO connector, a conventional 10 mm high MT-RJ connector, a conventional 12.7 mm high LC duplex connector, or a conventional 9.3 mm high SC duplex connector. In some embodiments, the simplex MT connector is a simplex micro-MT connector having a height **710** in a range of about 3.8 mm to about 7.5 mm, such as a height of about 6.5 mm or a height of about 5.5 mm.

The simplex micro-MT connector **700** shown in FIG. **7D** has a width **712** smaller than that of a conventional 12.4 mm wide MPO connector, a conventional 9.2 mm wide MT-RJ connector, a conventional 12.8 mm wide LC duplex connector, or a conventional 24.2 mm wide SC duplex connector. In some embodiments, the MT simplex connector is a simplex micro-MT connector having a width **712** in a range of about 6.4 mm to about 9.1 mm, such as a width of about 8.8 mm or a width of about 8.5 mm.

FIG. **7E** shows a side view of the simplex micro-MT connector **700** having a length **714** smaller than that of a conventional 25.7 mm long MPO connector, a conventional 27.3 mm long LC duplex connector, or a conventional 25.2 mm long SC duplex connector. In some embodiments, the simplex MT connector is a simplex micro-MT connector having a length **714** in a range of about 17.0 mm to about 25.1 mm, such as a length of about 23.2 mm or a length of about 21.3 mm.

FIG. **7F** shows one embodiment of the simplex micro-MT connector **700** according to aspects disclosed herein. The simplex micro-MT connector **700** may include a housing **736** defining a longitudinal passage therethrough from a proximal end **736a** to a distal end **736b** and including a latch hole **736c** designed to couple with a latch arm (not shown) from an adaptor (not shown).

The simplex micro-MT connector **700** may include a latch body **730** defining a first longitudinal passage therethrough from a proximal end **730a** to a distal end **730b**, a second longitudinal passage therethrough from a middle **730c** to the distal end **730b**. The latch body **730** comprises a bottom **730d**, a hook hole **732** to couple with a pull-tab (not shown), and a latch hole **730e** to couple with a latch arm (not shown) from an adaptor (not shown). The latch body is movably

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coupled with the housing 736 wherein at least part of the housing is movably within at least the first longitudinal passage of the latch body 730.

The simplex micro-MT connector 700 may comprise a latch spring 734 defining a longitudinal passage there-
through and comprising arms 734a, lower parts 734b, and
upper parts 734c, and a top part 734d. The latch spring is
coupled with the distant end 730b and the bottom 730d of
the latch body 730. The latch spring is movably coupled with
the housing 736 wherein at least part of the housing is
movably within the longitudinal passage of the latch spring
734.

The simplex micro-MT connector 700 may comprise a
ferrule 738 extending longitudinally from a proximal end
746 to a distal end 748 and disposed at least partially within
the longitudinal passage of the housing 736 with the prox-
imal end protruding externally through the passage defined
by the latch body 730 and the housing.

The simplex micro-MT connector 700 may comprise a
ferrule spring 740 defining a longitudinal and movable
passage therethrough from a proximal end 740a to a distal
end 740b. The ferrule spring 740 is at least partially within
the longitudinal passage of the housing 736.

The simplex micro-MT connector 700 may comprise a
spring push 742 defining a longitudinal passage there-
through from a proximal end 750 to a distal end 752 and
comprising a flange 742a. The spring push 742 is at least
partially within the housing 736.

The simplex micro-MT connector 700 may comprise a
cable boot 754 defining a longitudinal passage therethrough
from a proximal end 754a to a distal end 754b. The proximal
end 754a of the cable boot 754 is coupled with the distal end
752 of the spring push 742.

As shown in FIG. 8A, the simplex micro-MT connector
700 may comprise a pull-tab 800 having a distal end 802 and
a proximal end 804. The proximal end 804 may include a
latch hook 806 uniquely shaped so as to engage with the
unique profile of the latch body 730 that is integral to the
simplex micro-MT connector 700. As shown in FIG. 8B, the
proximal end 804 is disposed within the second longitudinal
passage of the latch body 730. The pull-tab hook 806 snaps
with the hook hole 732 so that the pull-tab 800 is coupled to
the simplex micro-MT connector 700. The pull-tab 800 can
be removably coupled to the simplex micro-MT connector
700. The pull-tab 800 may be coupled to the simplex
micro-MT connector 700 by means other than by snapping
into position, such as sliding or hooking into position. The
pull-tab 800 may alternatively be an integral component of
the simplex micro-MT connector 700.

FIG. 9A shows the configuration of the simplex micro-
MT connector 700 when it is coupled with an adaptor (not
shown). In this configuration, the latch spring 734 is not
compressed and therefore not actuated. The lower part 734b
of the latch spring 734 is in contact with the latch body 730
distal end 730b, and the top part 734d of the latch spring 734
is in contact with the flange 742a from the spring push 742.
Most importantly, in this configuration, the latch hole 736c
from the housing 736 and the latch hole 730e from the latch
body 730 are disposed vertically, one above each other, to
generate a large cavity where a latch from an adaptor can
couple to fasten the connector to the adaptor.

FIG. 9B shows the configuration of the simplex micro-
MT connector 700 when it is not coupled with an adaptor
(not shown). In this configuration, the latch spring 734 is
compressed and therefore actuated. The latch body 730 is
pulled back manually or by the pull-tab 800 (not shown)
toward the distal end of the housing 736 and compresses the

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latch spring 734 so its lower part 734b, middle part 734c and
top 734d are in contact with both the distal end 730b of the
latch body 730 and the flange 742a of the push spring 742.
In this configuration, the latch hole 736c from the housing
736 and the latch hole 730e from the latch body 730 are not
disposed vertically, one above each other, and do not gener-
ate a large enough cavity where a latch from the adaptor
can couple to fasten the connector to the adaptor micro-MT
duplex connector.

Various embodiments disclosed herein are configured for
use as a mechanical transfer (MT) connector, such as a
duplex micro-MT (MMT) connector 1000 as shown in FIG.
10A. In some embodiments, the duplex micro-MT connector
1000 has a height 1010 in a range of about 7.8 mm to about
13.8 mm, such as a height of about 12.8 mm or a height of
about 11.8 mm. In some embodiments, the duplex micro-
MT connector 1000 has a width 1012 in a range of about 6.4
mm to about 9.1 mm, such as a width of about 8.8 mm or
a width of about 8.5 mm.

FIG. 10B shows a side view of the duplex micro-MT
connector 1000. In some embodiments, the duplex micro-
MT connector 1000 has a length 1014 in a range of about
19.4 mm to about 29.0 mm, such as a length of about 27.0
mm or a length of about 25.0 mm.

FIG. 10C shows one embodiment of the duplex micro-
MT connector 1000 according to aspects disclosed herein.
The duplex micro-MT connector 1000 may include a first
housing and a second housing 1036 each defining a longi-
tudinal passage therethrough from a proximal end 1036a to
a distal end 1036b and including a latch hole 1036c designed
to couple with a latch arm (not shown) from an adaptor (not
shown).

The duplex micro-MT connector 1000 may include a
latch body 1030 defining a first longitudinal passage there-
through from a proximal end 1030a to a distal end 1030b, a
second longitudinal passage therethrough from a middle
1030c to the distal end 1030b. The latch body 1030 com-
prises a bottom 1030d, two hook holes 1032 to couple with
a pull-tab (not shown), and two latch holes 1030e to couple
with a latch arm (not shown) from an adaptor (not shown).
The latch body 1030 is movably coupled with the housings
1036 wherein at least part of the housing is movably within
at least the first longitudinal passage of the latch body 1030.

The duplex micro-MT connector 1000 may comprise a latch
spring 1034 defining a longitudinal passage therethrough
and comprising arms lower parts 1034b, upper parts 1034c,
and a top part 1034d. The latch spring 1034 is coupled with
the distal end 1030b and the bottom 1030d of the latch body
1030. The latch is movably coupled with the housings 1036
wherein at least part of the housings is movably within the
longitudinal passage of the latch spring 1034.

The duplex micro-MT connector 1000 may comprise two
ferrules 1038 each extending longitudinally from a proximal
end 1046 to a distal end 1048 and disposed at least partially
within the longitudinal passage of one of the housings 1036
with the proximal end 1046 protruding externally to the
passage.

The duplex micro-MT connector 1000 may comprise two
ferrule springs 1040 defining a longitudinal and movable
passage therethrough from a proximal end 1040a to a distal
end 1040b. The ferrule springs 1040 are at least partially
within a longitudinal passage of the housing 1036.

The duplex micro-MT connector 1000 may comprise a
spring separator 1056 defining two longitudinal passages
therethrough and comprising an elongated part extending
from a proximal end 1056a to a distal end 1056b.

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The duplex micro-MT connector **1000** may comprise a spring push **1042** defining a longitudinal passage there-through from a proximal end **1050** to a distal end **1052** and comprising a flange **1042a**. The spring push **1042** is at least partially within the housing **1036**.

The duplex micro-MT connector **1000** may comprise a cable boot **1054** defining a longitudinal passage there-through from a proximal end **1054a** to a distal end **1054b**. The proximal end **1054a** of the cable boot **1054** is coupled with the distal end **1052** of the spring push **1042**.

As shown in FIG. 11A, the duplex micro-MT connector **1000** may comprise a pull-tab **1100** having a distal end **1102** and a proximal end **1104**. The proximal end **1104** may include two latch hooks **1106a** and **1106b** uniquely shaped so as to engage with the unique profile of latch body **1030** from the duplex micro-MT connector **1000**. As shown in FIG. 11B, proximal end **1104** is disposed within the second longitudinal passage of the latch body **1030**. The pull-tab hooks **1106a** and **1106b** then snaps with the hook holes **1032** so that the pull-tab **1100** is coupled to the duplex micro-MT connector **1000**.

Various embodiments disclosed herein are configured for use as a mechanical transfer (MT) connector, such as a quad micro-MT (MMT) connector **1200** as shown in FIG. 12A. In some embodiments, the quad micro-MT connector **1200** has a height **1210** in a range of about 7.8 mm to about 13.8 mm, or such as a height of about 7.8 mm to about 12.8 mm or a height of about 7.8 mm to about 11.8 mm. In some embodiments, the quad micro-MT connector **1200** has a width **1212** in a range of about 12.8 mm to about 21.2 mm, such as a width of about 20.2 mm or a width of about 19.2 mm.

FIG. 12B shows a side view of the quad micro-MT connector **1200**. In some embodiments, the quad micro-MT connector **1200** has a length **1214** in a range of about 20.2 mm to about 32.0 mm, such as a length of about 31.0 mm or a length of about 30.0 mm.

FIG. 12C shows one embodiment of quad micro-MT connector **1200** according to aspects disclosed herein. The quad micro-MT connector **1200** may include four housings **1236** each defining a longitudinal passage therethrough from a proximal end **1236a** to a distal end **1236b** and including a latch hole **1236c** designed to couple with a latch arm (not shown) from an adaptor (not shown).

The quad micro-MT connector **1200** may include a latch body **1230** defining a first longitudinal passage therethrough from a proximal end **1230a** to a distal end **1230b**, a second longitudinal passage therethrough from a middle **1230c** to the distal end **1230b**. The latch body **1230** comprises a bottom **1230d**, two hook holes **1232** to couple with a pull-tab (not shown), and four latch holes **1230e** to couple with a latch arm (not shown) from an adaptor (not shown). The latch body is movably coupled with the housings **1236** wherein at least part of the housings is movably within at least the first longitudinal passage of the latch body.

The quad micro-MT connector **1200** may comprise two latch springs **1234** each defining a longitudinal passage therethrough and comprising arms **1234a**, lower parts **1234b**, upper parts **1234c**, and a top part **1234d**. The latch springs are coupled with the distal end **1230b** of the latch body **1230**. The latch springs are movably coupled with the housings **1236** wherein at least part of the housings is movably within the longitudinal passage of the latch springs **1234**.

The quad micro-MT connector **1200** may comprise four ferrules **1238** each extending longitudinally from a proximal end **1246** to a distal end **1248** and disposed at least partially

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within the longitudinal passage of one housing **1236** with the proximal end **1246** protruding externally to the passage.

The quad micro-MT connector **1200** may comprise four ferrule springs **1240** defining a longitudinal and movable passage therethrough from a proximal end **1240a** to a distal end **1240b**. The ferrule springs **1240** are at least partially within the longitudinal passage of housings **1236**.

The quad micro-MT connector **1200** may comprise a spring separator **1256** defining four longitudinal passages therethrough and comprising two elongated parts extending from a proximal end **1256a** to a distal end **1256b**.

The quad micro-MT connector **1200** may comprise a spring push **1242** defining a longitudinal passage there-through from a proximal end **1250** to a distal end **1252** and comprising a flange **1242a**. The spring push **1242** is at least partially within the housing **1236**.

The quad micro-MT connector **1200** may comprise a cable boot **1254** defining a longitudinal passage there-through from a proximal end **1254a** to a distal end **1254b**. The proximal end **1254a** of the cable boot **1254** is coupled with the distal end **1252** of the spring push **1242**.

As shown in FIG. 13A, the quad micro-MT connector **1200** may comprise a pull-tab **1300** having a distal end **1302** and a proximal end **1304**. The proximal end **1304** may include two latch hooks **1306a** and **1306b** uniquely shaped so as to engage with the unique profile of latch body **1230** from the quad micro-MT connector **1200**. As shown in FIG. 13B, proximal end is disposed within the second longitudinal passage of the latch body **1230**. The pull-tab hooks **1306a** and **1306b** then snap with the hook holes **1232** so that the pull-tab **1300** is coupled to the quad micro-MT connector **1200**.

Various embodiments disclosed herein are configured for use as an adaptor, such as a simplex micro-MT adaptor **1400** as shown in FIGS. 14A and 14B. FIG. 14A shows a top view of the simplex micro-MT adaptor **1400**, and FIG. 14B shows an end view of the simplex micro-MT adaptor **1400** without a flange. In some embodiments, the simplex micro-MT adaptor **1400** has a length **1402** in a range of about 22 mm to about 28 mm, such as a length of about 27 mm or a length of about 26 mm. In some embodiments, the simplex micro-MT adaptor **1400** has a height **1404** in a range of about 4 mm to about 10 mm, such as a height of about 9 mm or a height of about 8 mm. In some embodiments, the simplex micro-MT adaptor **1400** has a width **1406** in a range of about 7.2 mm to about 13.2 mm, such as a width of about 12.2 mm or a width of about 11.2 mm. In some embodiments, the simplex micro-MT adaptor **1400** has mounting handles **1407** and a height **1408** in a range of about 7 mm to about 13 mm, such as a height of about 12 mm or a height of about 11 mm.

Various embodiments disclosed herein are configured for use as an adaptor, such as a duplex micro-MT adaptor **1500** as shown in FIGS. 15A, 15B, and 15C. FIG. 15A shows a top view of the duplex micro-MT adaptor **1500**, FIG. 15B shows an end view of the duplex micro-MT adaptor **1500** without a flange, and FIG. 15C shows an end view of the duplex micro-MT adaptor **1500** with a flange. In some embodiments, the duplex micro-MT adaptor **1500** has a length **1502** in a range of about 22 mm to about 28 mm, such as a length of about 27 mm or a length of about 26 mm. In some embodiments, the duplex micro-MT adaptor **1500** has a height **1504** in a range of about 9 mm to about 15 mm, such as a height of about 14 mm or a height of about 13 mm. In some embodiments, the duplex micro-MT adaptor **1500** has a width **1506** in a range of about 7.2 mm to about 13.2 mm, such as a width of about 12.2 mm or a width of about 11.2 mm. In some embodiments, the duplex micro-MT adaptor

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1500 has mounting handles **1507** and a height **1508** in a range of about 12 mm to about 18 mm, such as a height of about 17 mm or a height of about 16 mm. In some embodiments, the duplex micro-MT adaptor **1500** has flanges **1509** and a height **1510** in a range of about 18 mm to about 24 mm, such as a height of about 23 mm or a height of about 22 mm.

Various embodiments disclosed herein are configured for use as an adaptor, such as a quad micro-MT adaptor **1600** as shown in FIGS. **16A**, **16B**, and **16C**. FIG. **16A** shows a top view of the quad micro-MT adaptor **1600**, FIG. **16B** shows an end view of the quad micro-MT adaptor **1600** without a flange, and FIG. **16C** shows an end view of the quad micro-MT adaptor **1600** with a flange. In some embodiments, the quad micro-MT adaptor **1600** has a length **1602** in a range of about 22 mm to about 28 mm, such as a length of about 27 mm or a length of about 26 mm. In some embodiments, the quad micro-MT adaptor **1600** has a height **1604** in a range of about 15.3 mm to about 21.3 mm, such as a height of about 20.3 mm or a height of about 19.3 mm. In some embodiments, the quad micro-MT adaptor **1600** has mounting handles **1607** and a width **1608** in a range of about 9 mm to about 15 mm, such as a height of about 14 mm or a height of about 13 mm. In some embodiments, the quad micro-MT adaptor **1600** has a height **1606** in a range of about 18.3 mm to about 24.3 mm, such as a height of about 23.3 mm or a width of about 22.3 mm. In some embodiments, the quad micro-MT adaptor **1600** has flanges **1609** and a height **1610** in a range of about 24.5 mm to about 30.5 mm, such as a height of about 29.5 mm or a height of about 28.5 mm.

FIG. **17A** shows an exploded view of the simplex micro-MT adaptor **1400**, and FIG. **17B** shows a perspective view of the simplex micro-MT adaptor **1400**. The simplex micro-MT adaptor **1400** comprises a housing **1402** defining a longitudinal passage therethrough, a proximal end **1404** and a distal end **1406**, wherein the proximal end **1404** is spaced from the distal end **1406**. The housing **1402** comprises at least one mounting tab **1408**, a first wall **1410**, a second wall **1412**, a third wall (not shown), and a fourth wall (not shown), wherein the first wall and the third wall are opposite to each other, the second wall and the fourth wall are opposite to each other, and the second wall **1412** comprises at least one adaptor housing hole **1414**. The adaptor **1400** includes at least one latch arm **1416** defined by a proximal end **1418** and a distal end **1420**, wherein the proximal end is spaced from the distal end. The proximal end **1418** and the distal end **1420** may each comprise a hook to couple with the adaptor **1400**, and the top side of the latch arm **1416** comprises a hook **1422** to couple with the adaptor housing hole **1414**.

FIG. **18A** shows an exploded view of the duplex micro-MT adaptor **1500**, and FIG. **18B** shows a perspective view of the duplex micro-MT adaptor **1500**. The duplex micro-MT adaptor **1500** comprises a housing **1502** defining a longitudinal passage therethrough, a proximal end **1504** and a distal end **1506**, wherein the proximal end **1504** is spaced from the distal end **1506**. The housing **1502** comprise at least one mounting tab **1508**, a first wall **1510**, a second wall **1512**, a third wall (not shown), and a fourth wall (not shown), wherein the first wall and the third wall are opposite to each other, the second and fourth walls are opposite to each other, and the second wall **1512** comprises two adaptor housing holes **1514**. The duplex micro-MT adaptor **1500** includes at least one latch arm **1516** defined by a proximal end **1518** and a distal end **1520**, wherein the proximal end is spaced from the distal end. The proximal end **1518** and the distal end **1520** may each comprise a hook to couple with the

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adaptor **1500**, and a top side of each latch arm **1516** comprises a hook **1522** to couple with an adaptor housing hole **1514**.

FIG. **19A** shows an exploded view of the quad micro-MT adaptor **1600**, and FIG. **19B** shows a perspective view of the quad micro-MT adaptor **1600**. The quad micro-MT adaptor **1600** comprises a housing **1602** defining a longitudinal passage therethrough, a proximal end **1604** and a distal end **1606**, wherein the proximal end **1604** is spaced from the distal end **1606**. The housing **1602** comprises at least one mounting tab **1608** that attaches to the housing, a first wall **1610**, a second wall **1612**, a third wall (not shown), and a fourth wall (not shown), wherein the first wall and the third wall are opposite to each other, the second and fourth walls are opposite to each other, the second wall **1612** comprises two adaptor housing holes **1614**, and the third wall (not shown) comprises two holes. The quad micro-MT adaptor **1600** includes four latch arms **1616** defined by a proximal end **1618** and a distal end **1620**, wherein the proximal end is spaced from the distal end. The proximal end **1618** and the distal end **1620** may each comprise a hook to couple with the adaptor **1600**, and each latch arm **1616** comprises a hook **1622** to couple with an adaptor housing hole **1614**.

FIG. **20A** shows an exploded view of a duplex micro-MT adaptor **1500** coupled with a duplex micro-MT connector **1000**, a first simplex micro-MT connector **700**, and a second simplex micro-MT connector **700**. FIG. **20B** shows a perspective view of the duplex micro-MT adaptor **1500** coupled with a duplex micro-MT connector **1000**, a first simplex micro-MT connector **700**, and a second simplex micro-MT connector **700**.

FIG. **21A** shows an exploded view of a quad micro-MT adaptor **1600** coupled with a first simplex micro-MT connector **700**, a second simplex micro-MT connector **700**, a duplex micro-MT connector **1000**, and a quad micro-MT **1200** connector. FIG. **21B** shows a perspective view of the quad micro-MT adaptor **1600** coupled with a first simplex micro-MT connector **700**, a second simplex micro-MT connector **700**, a duplex micro-MT connector **1000**, and a quad micro-MT **1200** connector.

What is claimed is:

1. A breakout method in a high density network panel application, the breakout method comprising:

plugging an optical fiber connector terminating a single optical fiber cable into a first end portion of an optical fiber adapter such that a respective group of optical fibers of the optical fiber connector are received in each of four channels of the optical fiber adapter;

plugging a plurality of individual optical fiber connectors into a second end portion of the optical fiber adapter such that one of the individual optical fiber connectors is received in each of the four channels of the optical fiber adapter, each of the plurality of individual optical fiber connectors terminating an individual optical fiber cable;

wherein said plugging the optical fiber connector and said plugging the individual optical fiber connectors makes optical connections between the individual optical fiber connectors and the groups of optical fibers in each of the four channels and thereby breaks out the single optical fiber cable into the plurality of individual cables;

wherein said plugging a plurality of individual optical fiber connectors into a second end portion of the optical fiber adapter latches the plurality of individual optical fiber connectors with the optical fiber adapter and such that each of the plurality of individual optical fiber

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connectors is configured to be individually unlatched from the optical fiber adapter for individual removal by pulling an externally accessible latch release backward away from the optical fiber adapter to displace the latch release in relation to a remainder of the respective individual optical fiber connector such that the individual optical fiber connector can be individually removed without removing any other individual optical fiber connector.

2. The breakout method of claim 1, wherein the optical fiber adapter comprises an adapter housing having first and second cross-sectional outer dimensions, the first and second cross-sectional outer dimensions being perpendicular, the first cross-sectional outer dimension being greater than the second cross-sectional outer dimension, the first cross-sectional outer dimension being in a range of from about 18.3 mm to about 24.3 mm.

3. The breakout method of claim 2, wherein the second cross-sectional outer dimension is in a range of from about 9 mm to about 15 mm.

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4. The breakout method of claim 1, wherein the optical fiber connector comprises four MT ferrules and each individual optical fiber connector comprises at least one MT ferrule.

5. The breakout method of claim 1, wherein said plugging the optical fiber connector and said plugging the individual optical fiber connectors makes four Tx-Rx connections between the optical fiber connector and the plurality of individual connectors.

6. The breakout method of claim 1, further comprising pulling an externally accessible latch release of one of the individual optical fiber connectors backward away from the optical fiber adapter to displace the latch release in relation to a remainder of said one of the individual optical fiber connectors to remove said one of the individual optical fiber connectors from the optical fiber adapter without removing any other individual optical fiber connector.

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